

Rookery Bay National Estuarine Research Reserve

SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

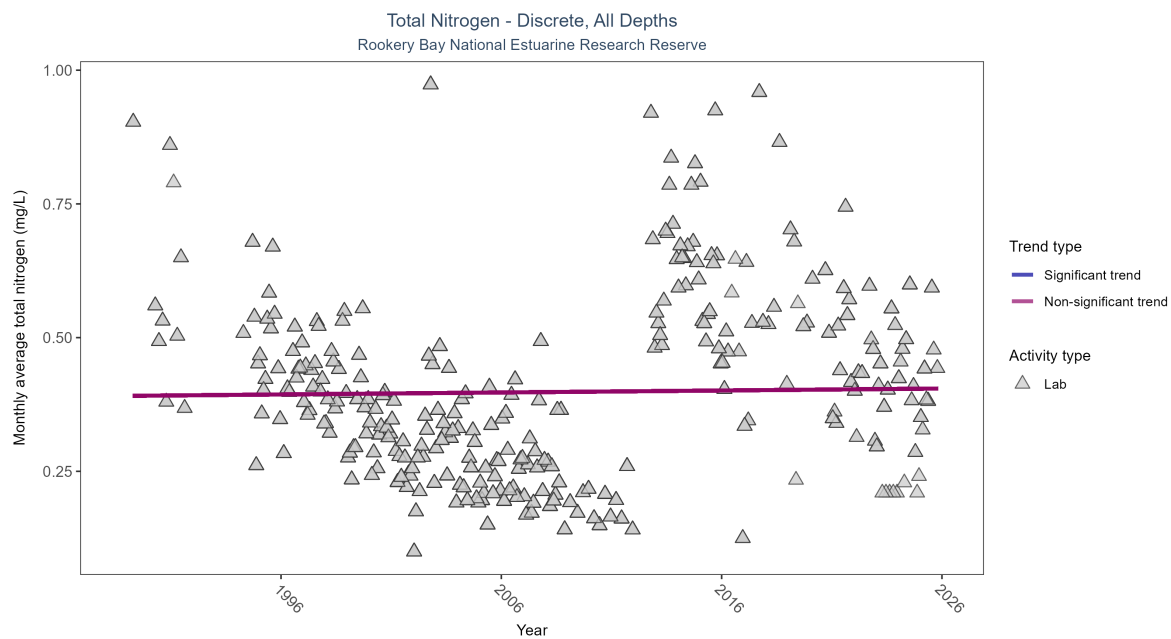


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	4746	35	1989 - 2025	0.31962	0.01298	0.39109	0.00037	0.8067

Total nitrogen showed no detectable trend between 1989 and 2025.

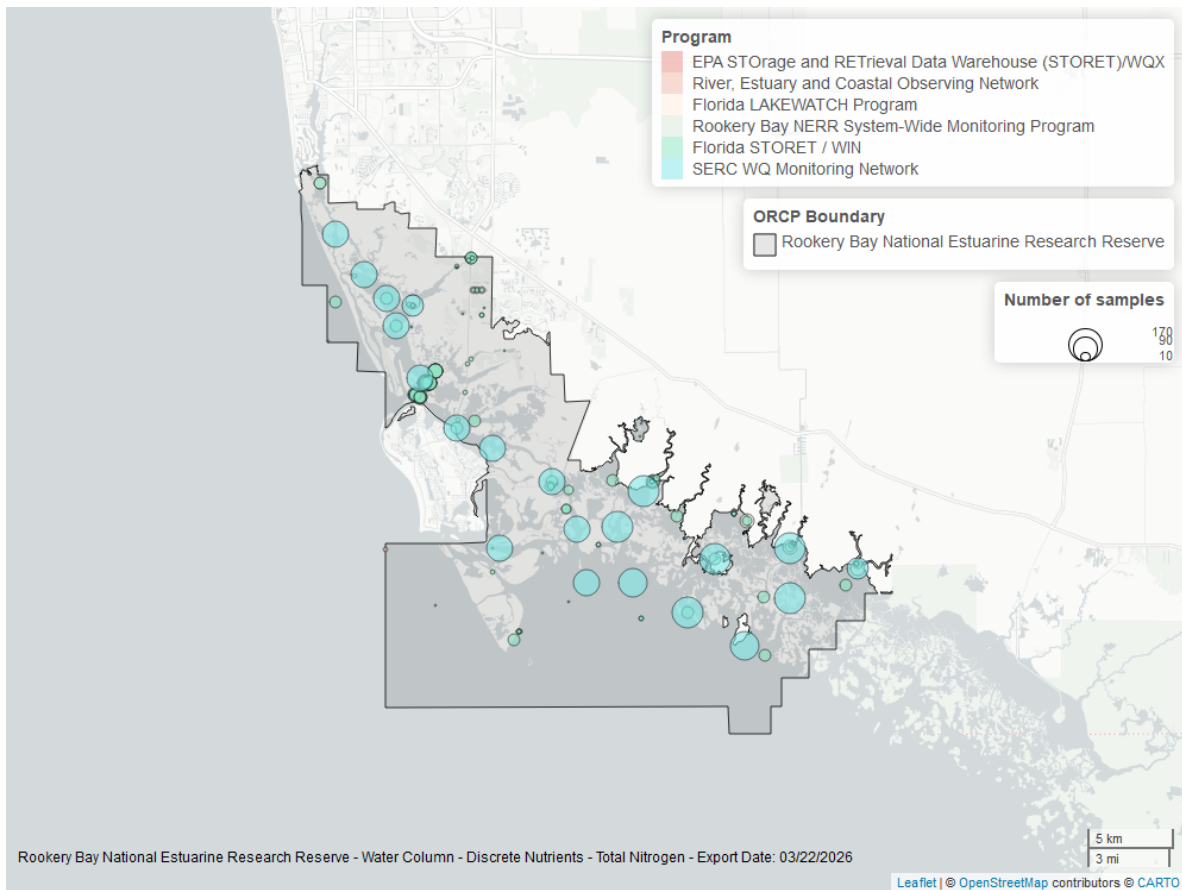


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

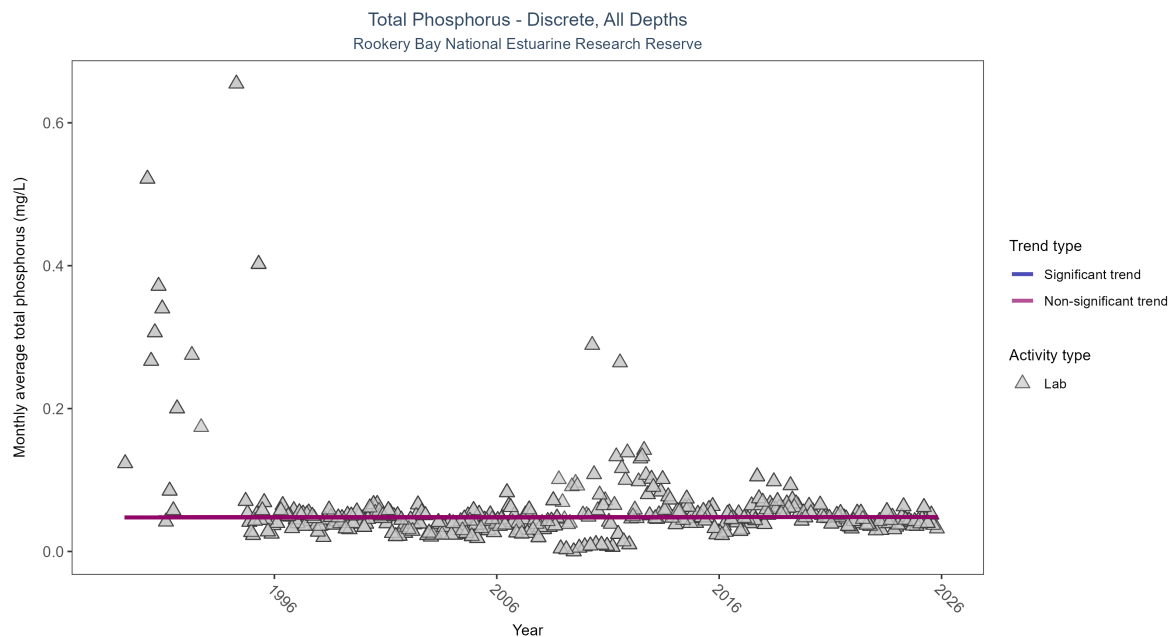


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	4881	36	1989 - 2025	0.04	0.00929	0.0477	0.00001	0.8776

Total phosphorus showed no detectable trend between 1989 and 2025.

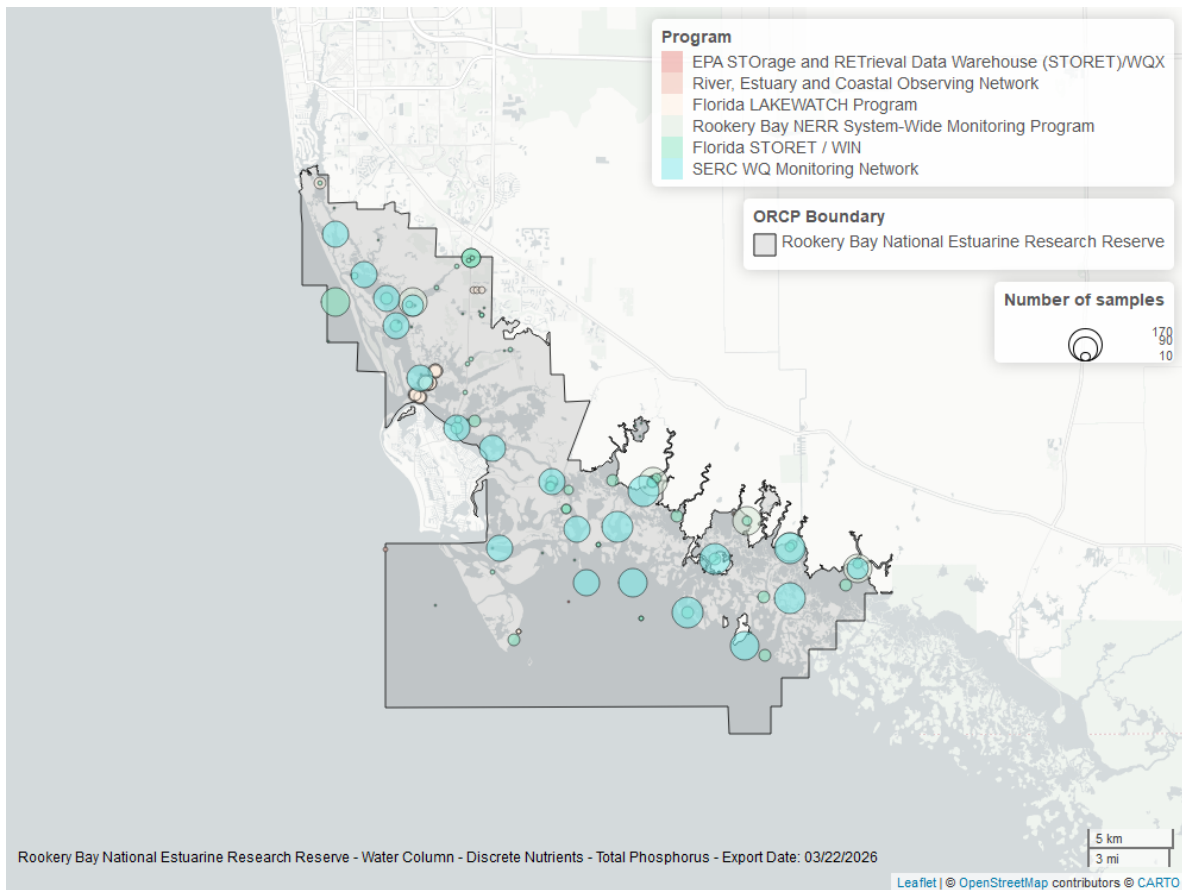


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

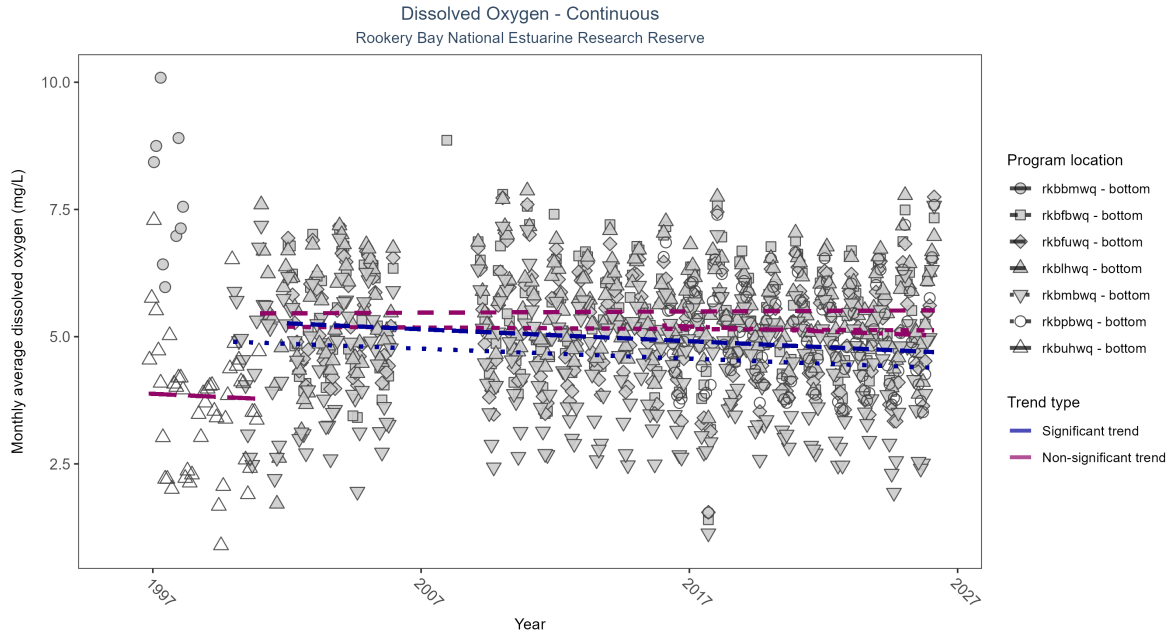


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwq	No significant trend	603909	23	2002 - 2026	5.4	-0.02	5.19	0	0.6197
rkbfuwq	Significantly decreasing trend	636515	22	2002 - 2026	5.1	-0.29	5.26	-0.02	0
rkmbmwq	Significantly decreasing trend	657119	24	2000 - 2026	4.4	-0.24	4.9	-0.02	0
rkbpbwq	No significant trend	332812	11	2016 - 2026	4.9	-0.14	5.22	-0.02	0.0762
rkbbmwq	Insufficient data to calculate trend	10441	2	1997 - 1998	7.2	-	-	-	-
rkblhwq	No significant trend	613216	23	2001 - 2026	5.6	0.03	5.46	0	0.5182
rkbuhwq	No significant trend	58164	5	1996 - 2000	3.6	0.01	3.9	-0.02	1

At two program locations, monthly average dissolved oxygen decreased by 0.02 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at four locations. There was insufficient data to fit a model for one location.

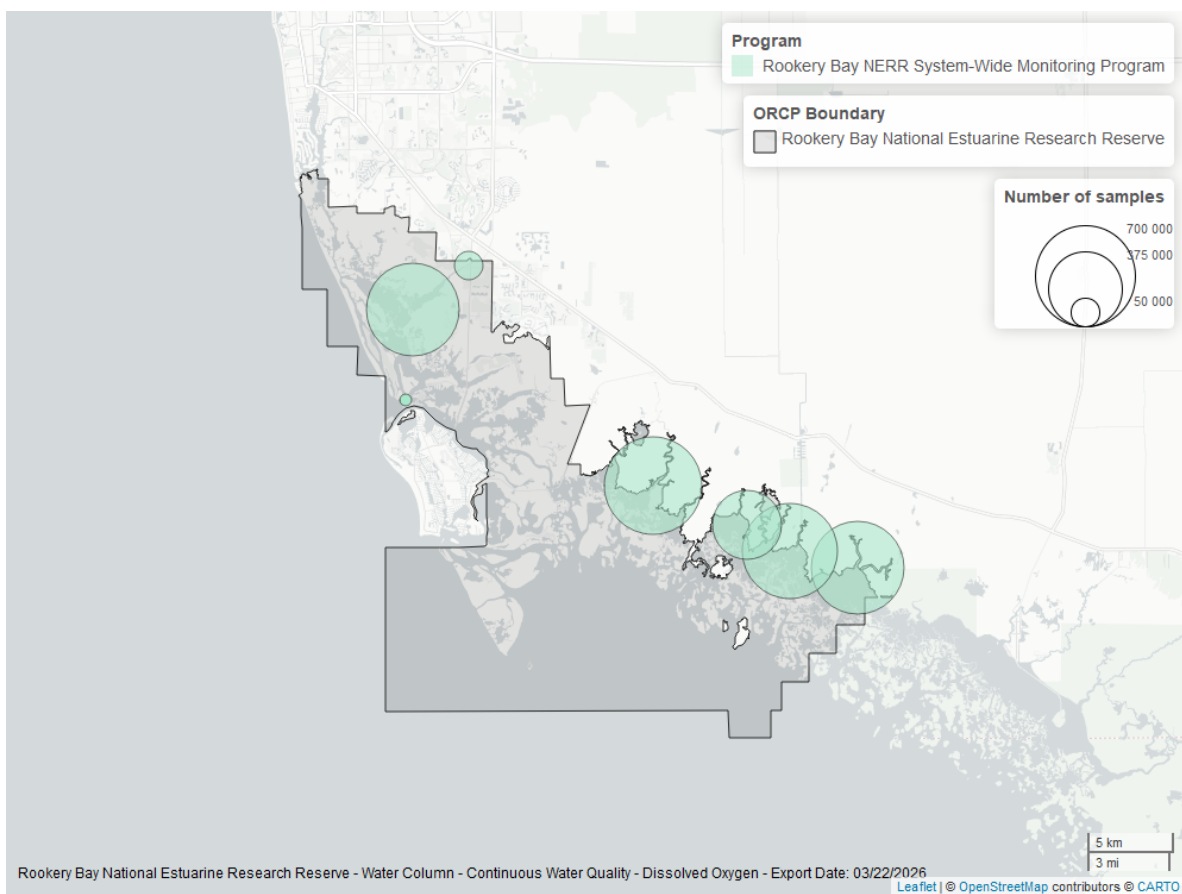


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

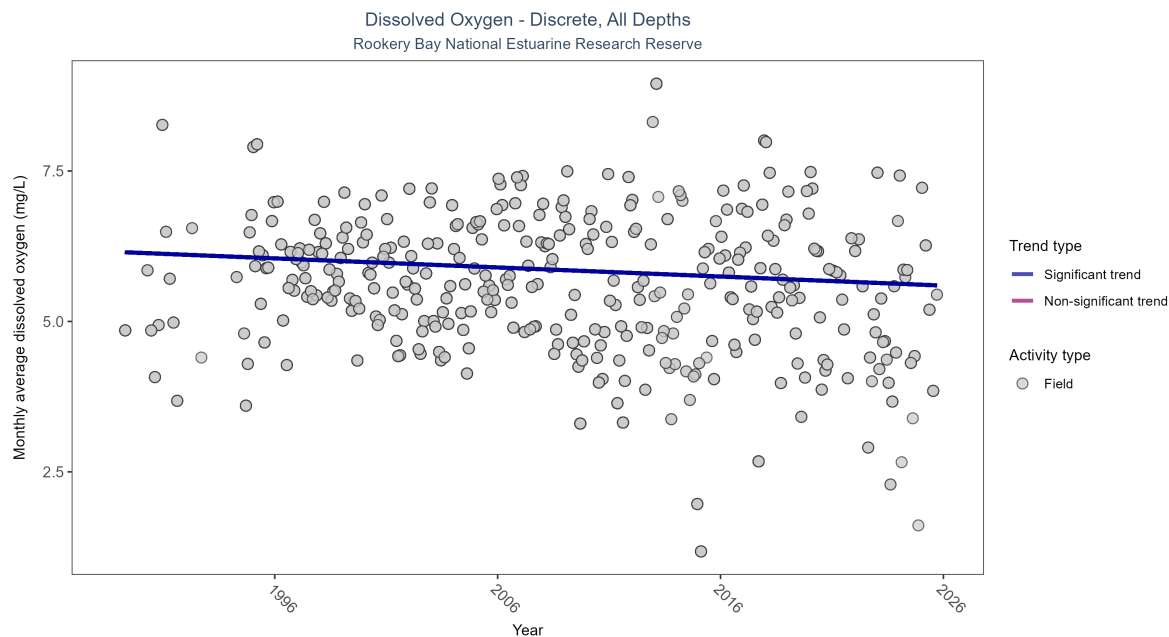


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	14164	36	1989 - 2025	5.8	-0.11754	6.15378	-0.01509	0.0014

Monthly average dissolved oxygen decreased by 0.02 mg/L per year.

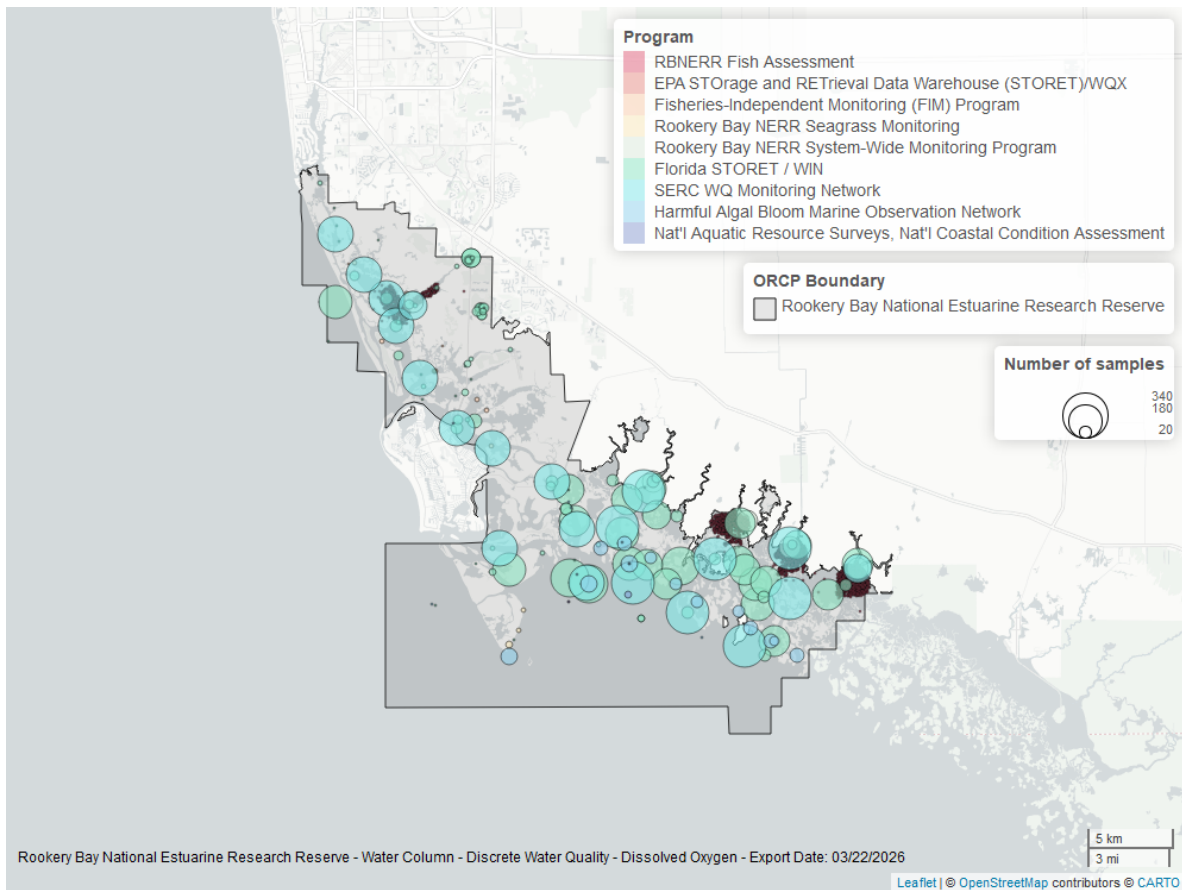


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

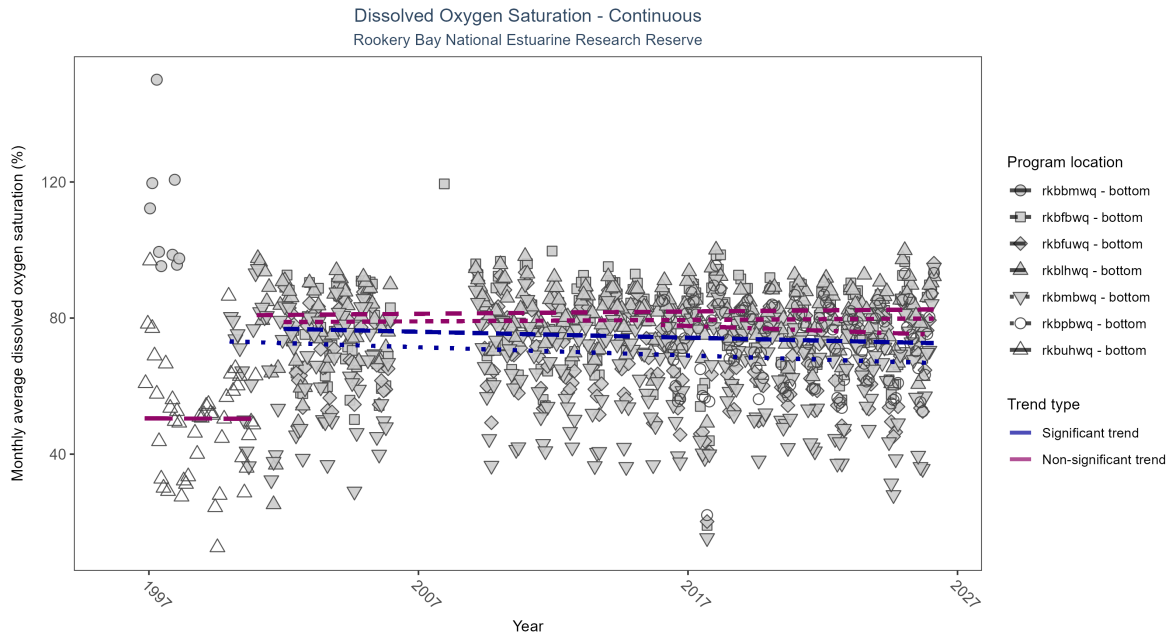


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwq	No significant trend	608215	23	2002 - 2026	79.0	0.04	78.81	0.04	0.4281
rkbfuwq	Significantly decreasing trend	636774	22	2002 - 2026	72.6	-0.2	76.84	-0.17	0
rkbbmwq	Significantly decreasing trend	663443	24	2000 - 2026	65.3	-0.21	73.13	-0.24	0
rkbpbwq	No significant trend	336249	11	2016 - 2026	72.7	-0.13	77.96	-0.28	0.0762
rkbbmwq	Insufficient data to calculate trend	10441	2	1997 - 1998	102.4	-	-	-	-
rkblhwq	No significant trend	625534	23	2001 - 2026	82.1	0.07	80.86	0.07	0.1302
rkbuhwq	No significant trend	58164	5	1996 - 2000	49.7	0	50.53	-0.02	1

At two program locations, monthly average dissolved oxygen saturation decreased by 0.17% per year at one site and by 0.24% per year at the other. No detectable change in monthly average dissolved oxygen saturation was observed at four locations. There was insufficient data to fit a model for one location.

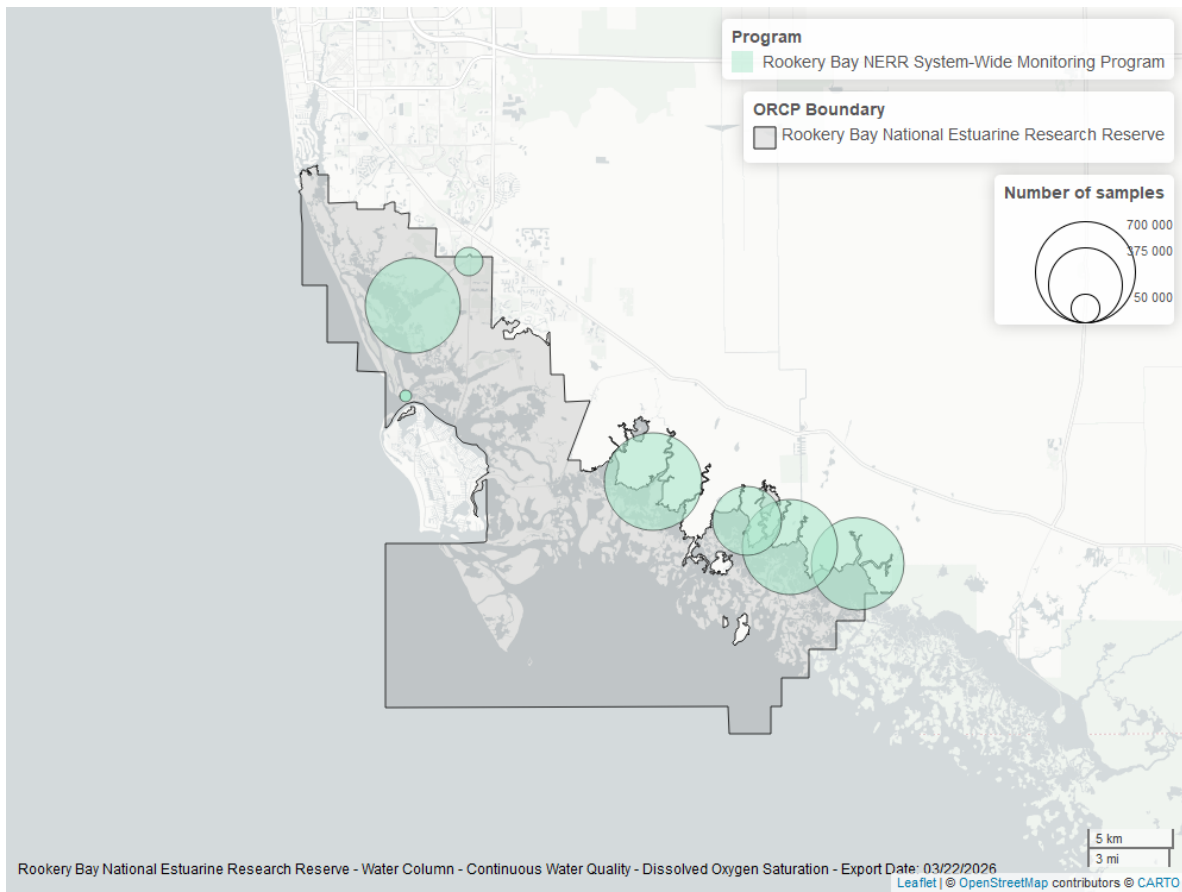


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

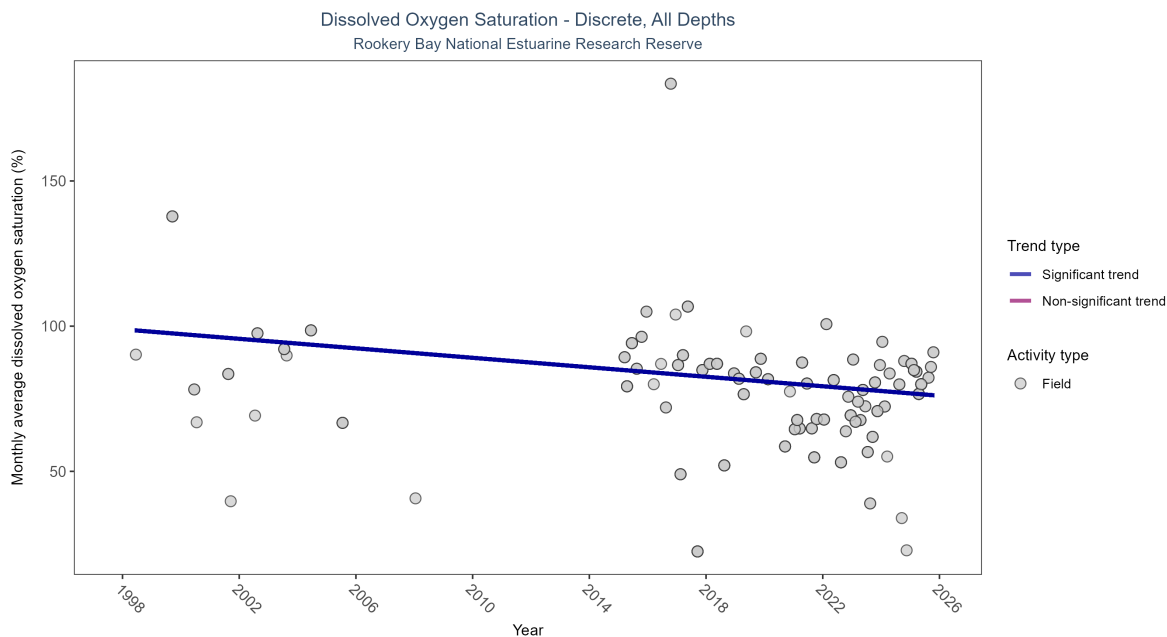


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	348	20	1998 - 2025	80	-0.24039	98.90328	-0.81667	0.0201

Monthly average dissolved oxygen saturation decreased by 0.82% per year.

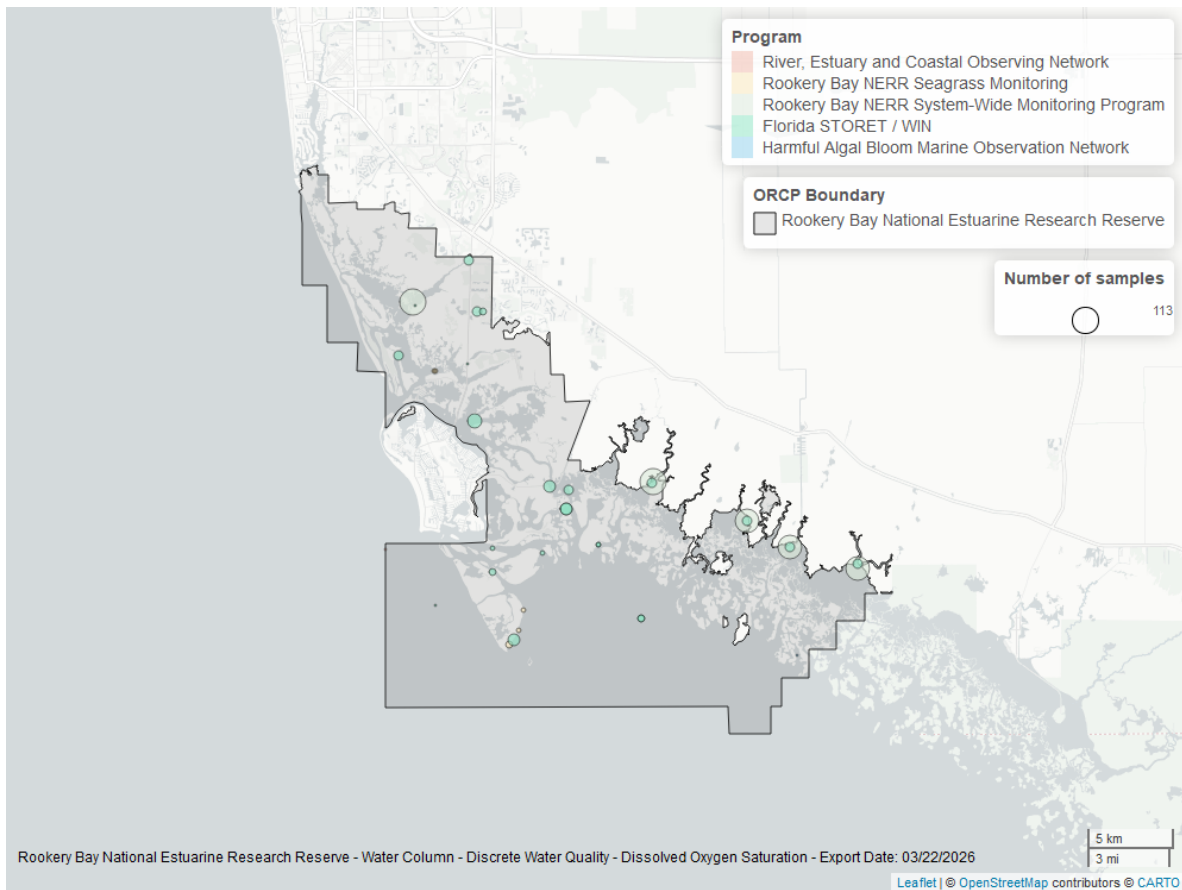


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

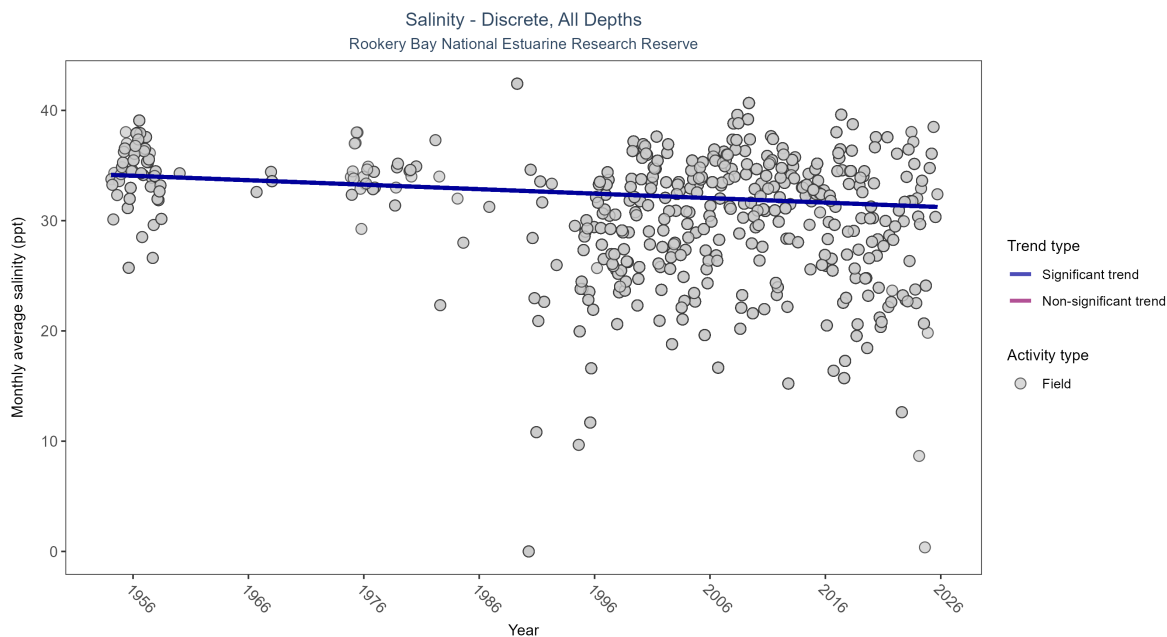


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Significantly decreasing trend	16960	54	1954 - 2025	32.705	-0.14316	34.15483	-0.04058	0

Monthly average salinity decreased by 0.04 ppt per year.

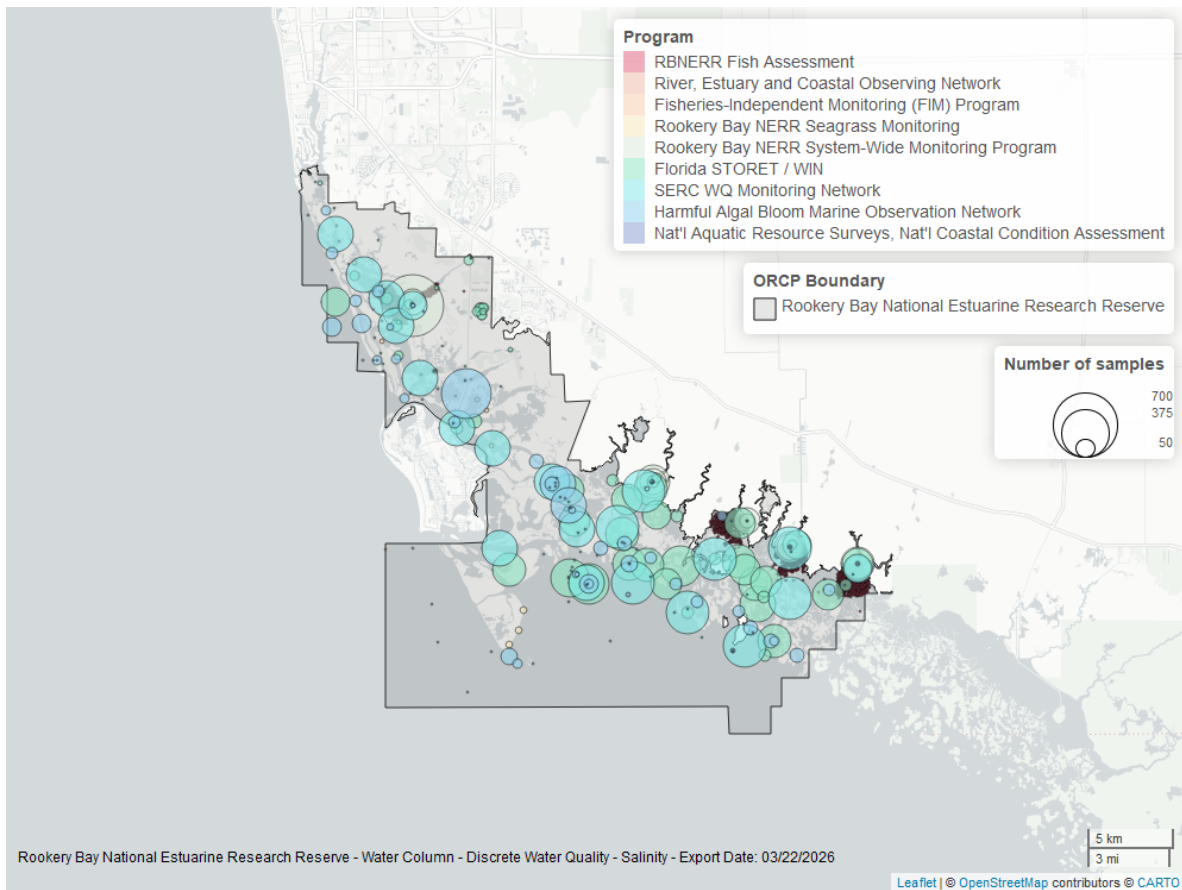


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

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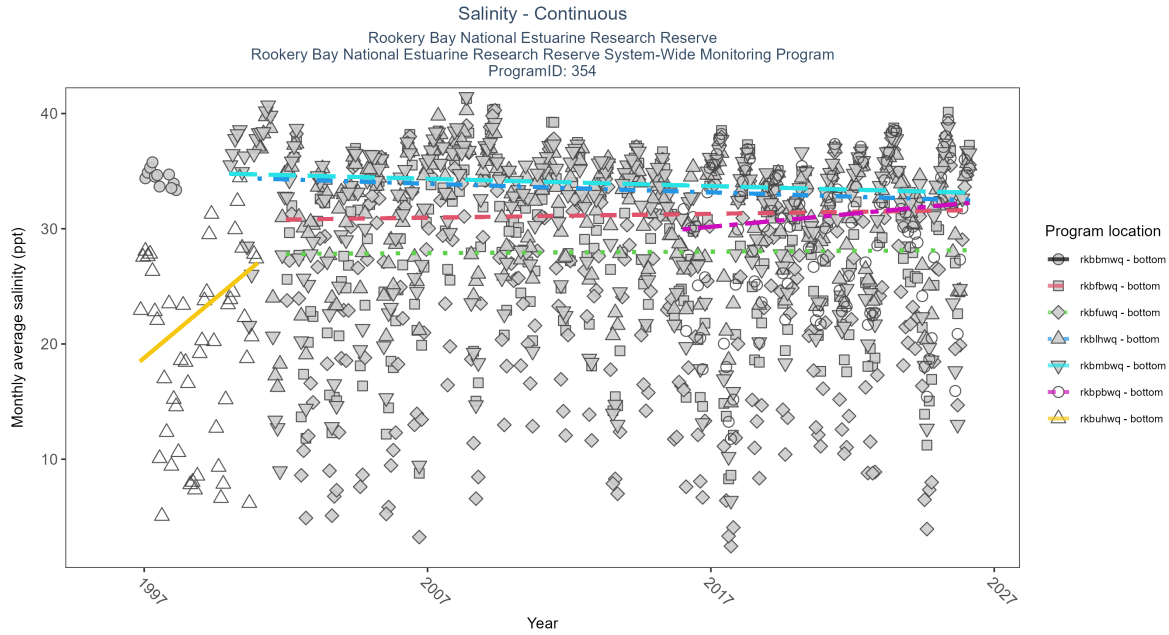


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results for All Stations - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfuwq	No significant trend	703182	25	2002 - 2026	29.9	0.04	30.8	0.03	0.3375
rkbfuwq	No significant trend	717069	25	2002 - 2026	26.3	0.01	27.82	0.01	0.7856
rkbbmwq	Significantly decreasing trend	736129	27	2000 - 2026	33.3	-0.1	34.78	-0.06	0.016
rkbpbwq	No significant trend	333127	11	2016 - 2026	30.5	0.14	29.94	0.23	0.0606
rkbbmwq	Insufficient data to calculate trend	12256	2	1997 - 1998	34.5	-	-	-	-
rkblhwq	Significantly decreasing trend	700367	26	2001 - 2026	33.2	-0.13	34.36	-0.08	0.0023
rkbuhwq	Significantly increasing trend	68446	5	1996 - 2000	21.1	0.31	16.74	2.06	0.0358

At two program locations, monthly average salinity increased by 0.31 ppt per year at one site and by 2.06 ppt per year at the other. At three program locations, monthly average salinity decreased between 0.06 and 0.44 ppt per year. No detectable change in monthly average salinity was observed at seven locations. There was insufficient data to fit a model for four locations.

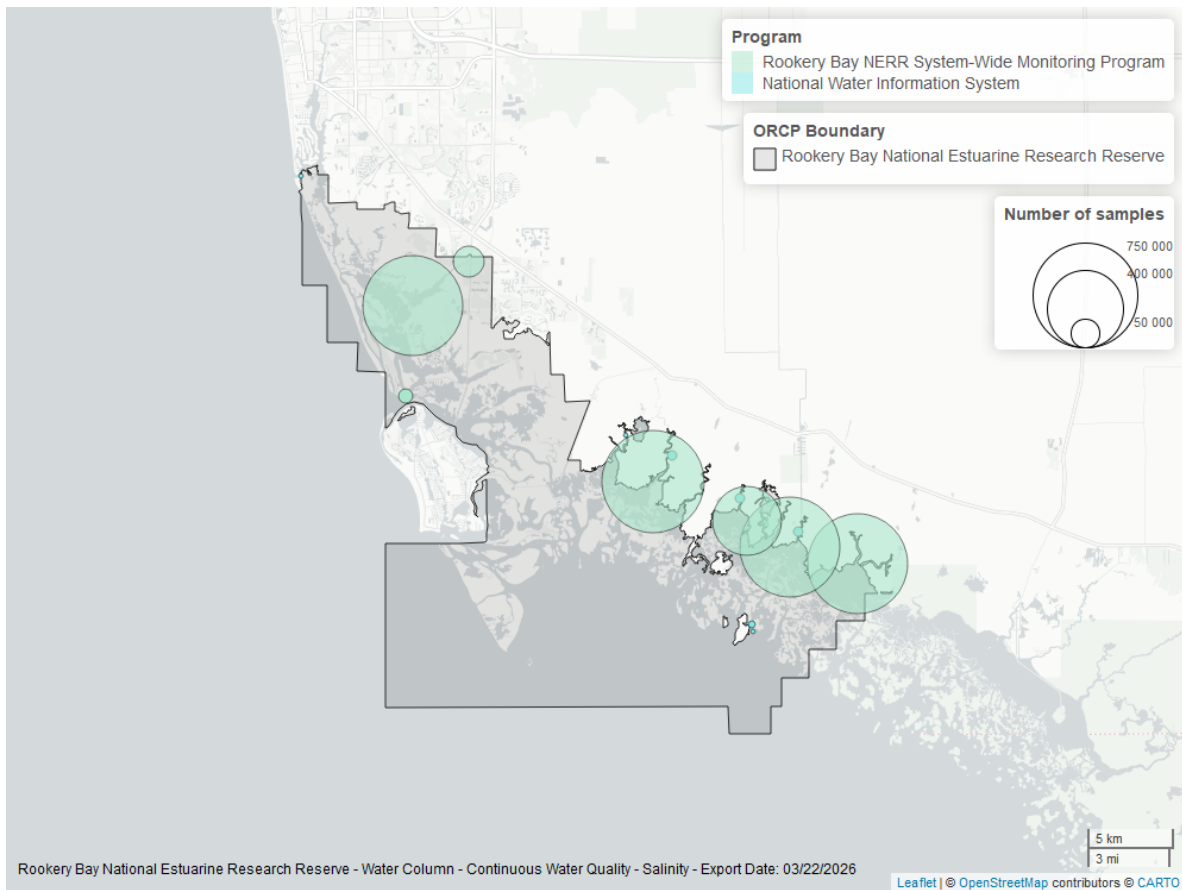


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

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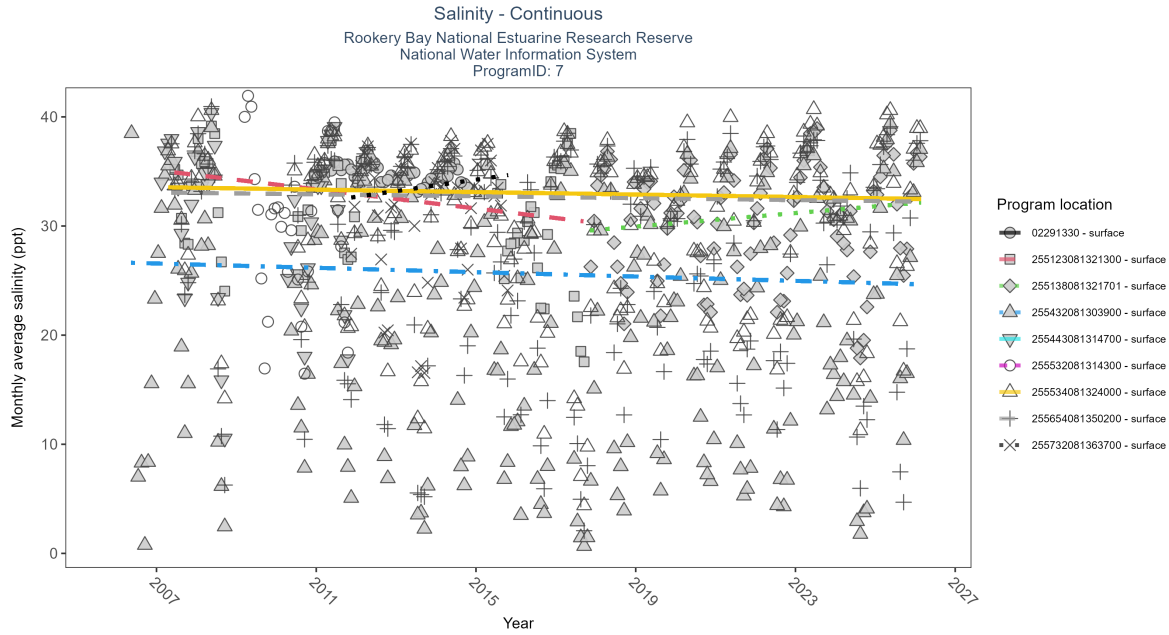


Figure 17: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results for All Stations - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	Significantly decreasing trend	1809	8	2007 - 2017	32	-0.23	35.11	-0.44	0.0182
255138081321701	Significantly increasing trend	2986	10	2017 - 2026	32	0.17	29.33	0.31	0.0294
255432081303900	No significant trend	6468	20	2006 - 2026	22	-0.08	26.67	-0.1	0.1135
255443081314700	Insufficient data to calculate trend	1465	4	2007 - 2011	32	-	-	-	-
255532081314300	Insufficient data to calculate trend	902	3	2009 - 2011	31	-	-	-	-
255534081324000	No significant trend	6162	19	2007 - 2026	32	-0.04	33.56	-0.06	0.3647
255654081350200	No significant trend	6183	19	2007 - 2026	32	-0.04	33.07	-0.04	0.365
255732081363700	No significant trend	1434	5	2011 - 2015	34	0.25	32.15	0.52	0.0955
02291330	Insufficient data to calculate trend	1697	4	2011 - 2014	35	-	-	-	-

At two program locations, monthly average salinity increased by 0.31 ppt per year at one site and by 2.06 ppt per year at the other. At three program locations, monthly average salinity decreased between 0.06 and 0.44 ppt per year. No detectable change in monthly average salinity was observed at seven locations. There was insufficient data to fit a model for four locations.

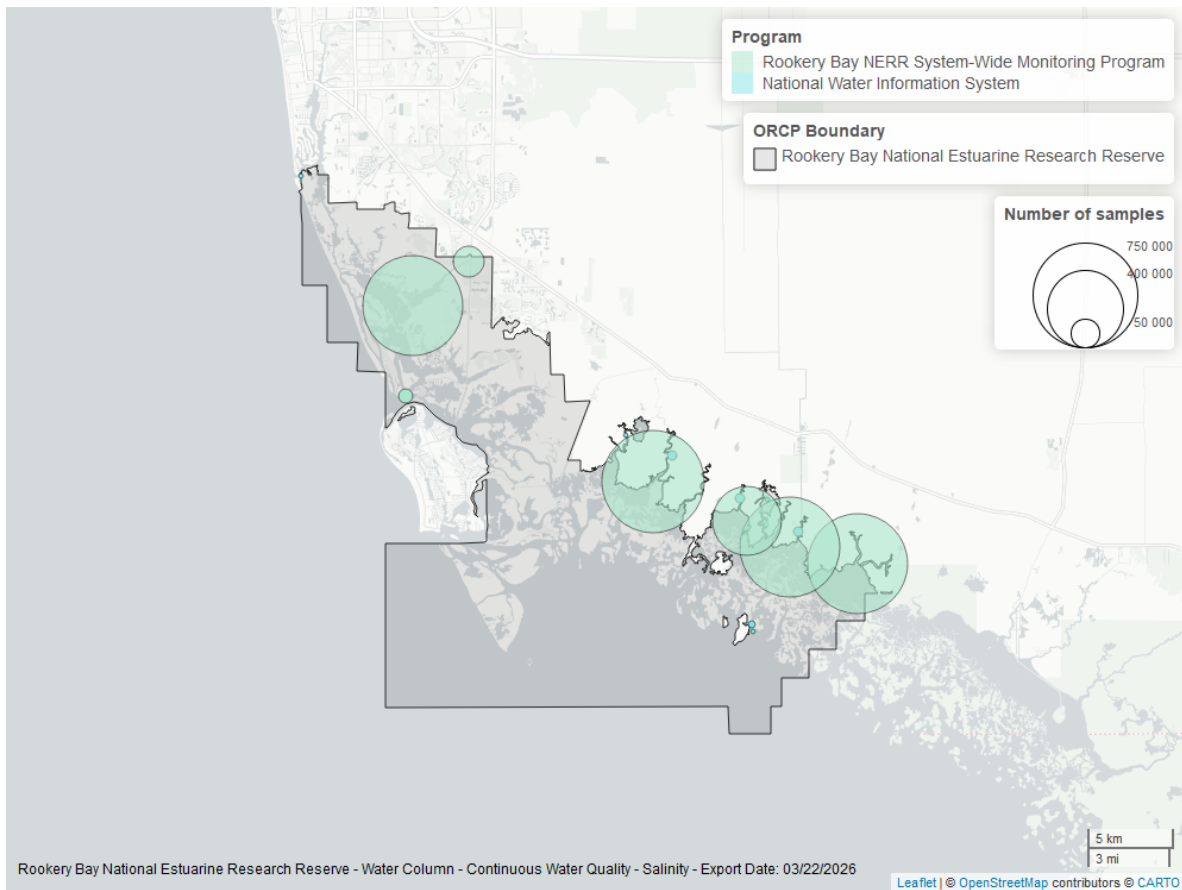


Figure 18: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

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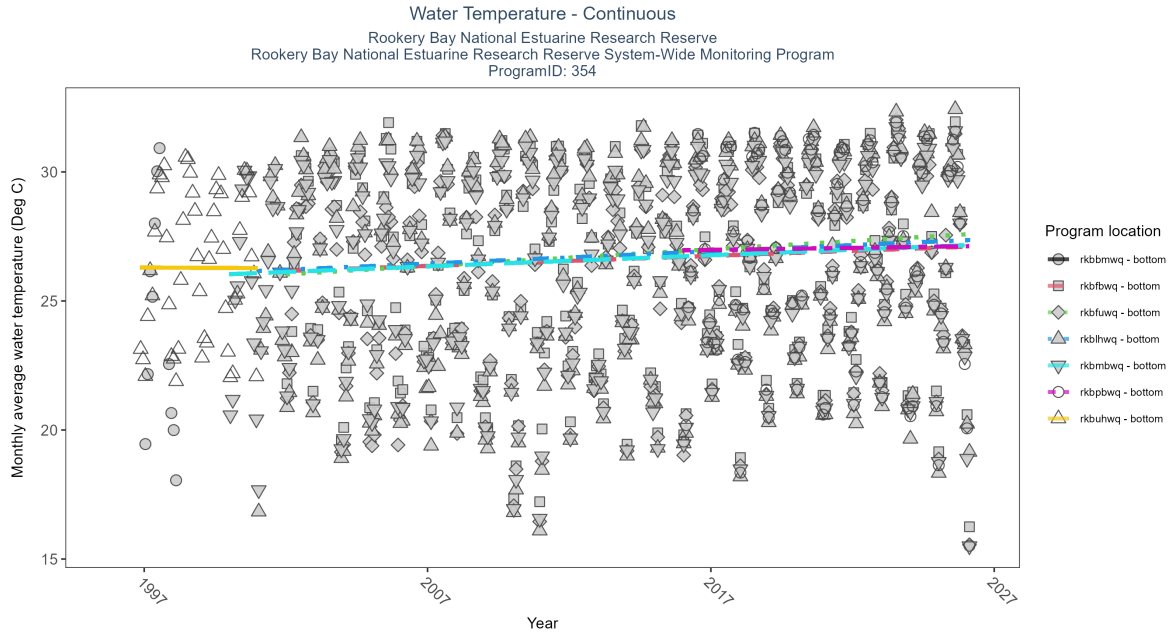


Figure 19: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: Seasonal Kendall-Tau Results for All Stations - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfuwq	Significantly increasing trend	719021	25	2002 - 2026	27.0	0.22	26.16	0.04	0
rkbfuwq	Significantly increasing trend	727501	25	2002 - 2026	26.9	0.26	26.03	0.06	0
rkbbmwq	Significantly increasing trend	762033	27	2000 - 2026	26.8	0.25	26.04	0.04	0
rkbpbwq	No significant trend	337833	11	2016 - 2026	27.4	0.05	26.96	0.02	0.5544
rkbbmwq	Insufficient data to calculate trend	12610	2	1997 - 1998	23.8	-	-	-	-
rkblhwq	Significantly increasing trend	731519	26	2001 - 2026	26.9	0.24	26.16	0.05	0
rkbuhwq	No significant trend	68971	5	1996 - 2000	26.8	-0.01	26.3	-0.01	1

At seven program locations, monthly average water temperature increased between 0.03 and 0.06°C per year. No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for four locations.

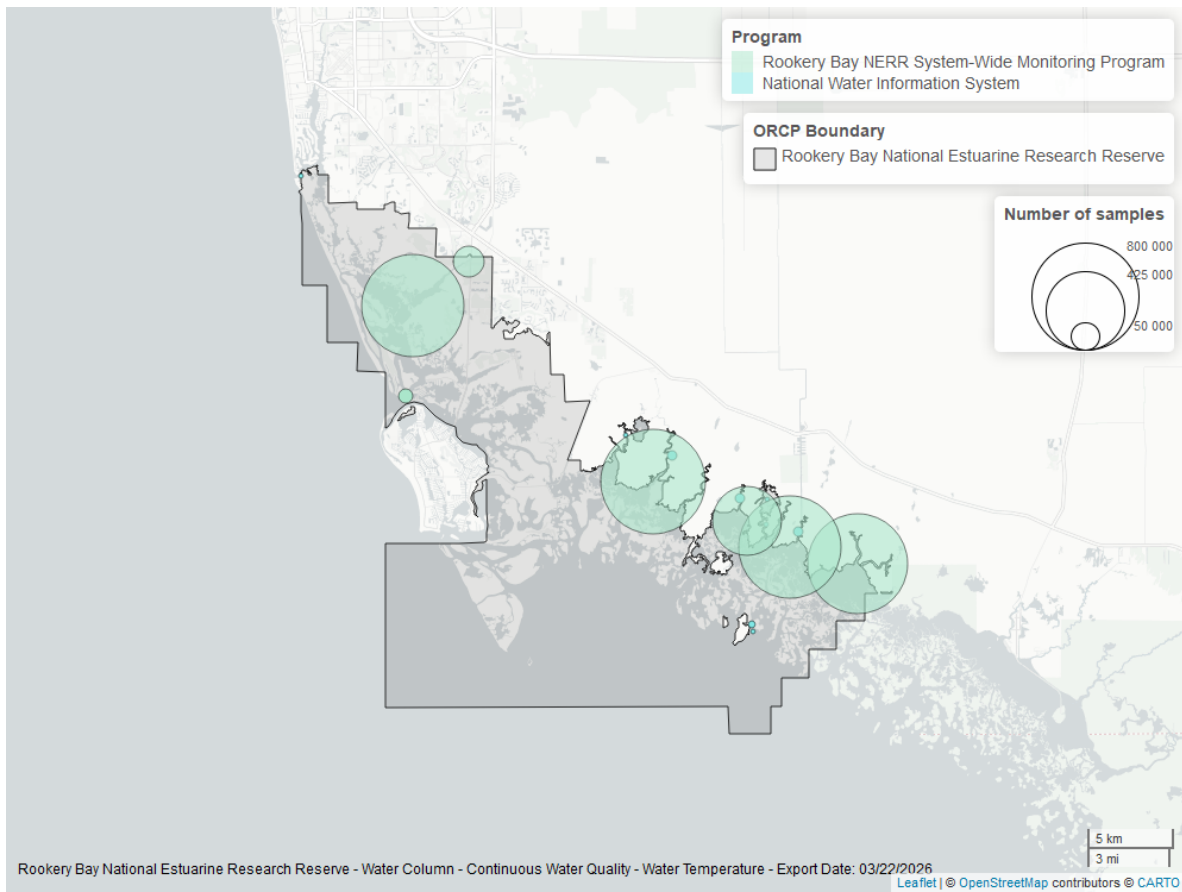


Figure 20: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

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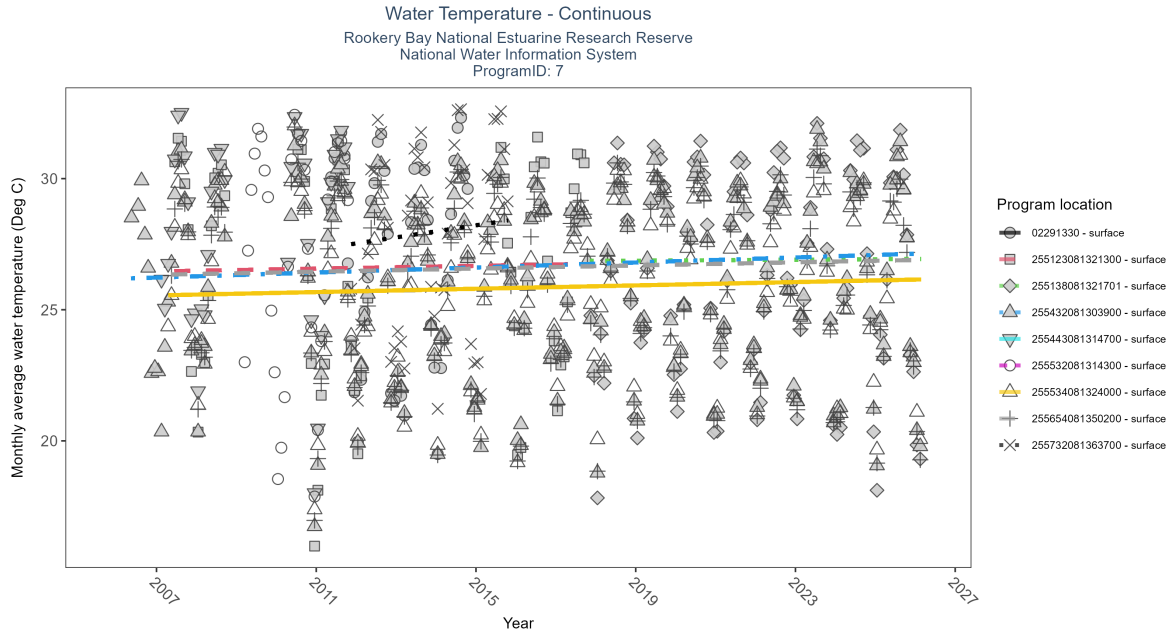


Figure 21: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results for All Stations - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	No significant trend	1818	8	2007 - 2017	27.3	0.1	26.46	0.03	0.3843
255138081321701	No significant trend	3005	10	2017 - 2026	27.0	0.02	26.85	0.01	0.9199
255432081303900	Significantly increasing trend	6527	20	2006 - 2026	27.0	0.17	26.18	0.05	0.0005
255443081314700	Insufficient data to calculate trend	2011	4	2007 - 2011	29.3	-	-	-	-
255532081314300	Insufficient data to calculate trend	906	3	2009 - 2011	29.3	-	-	-	-
255534081324000	Significantly increasing trend	6225	19	2007 - 2026	26.6	0.14	25.55	0.03	0.0069
255654081350200	Significantly increasing trend	6222	19	2007 - 2026	26.7	0.11	26.33	0.03	0.0327
255732081363700	No significant trend	1435	5	2011 - 2015	28.4	0.24	27.3	0.23	0.1149
02291330	Insufficient data to calculate trend	1901	4	2011 - 2014	28.3	-	-	-	-

At seven program locations, monthly average water temperature increased between 0.03 and 0.06°C per year. No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for four locations.

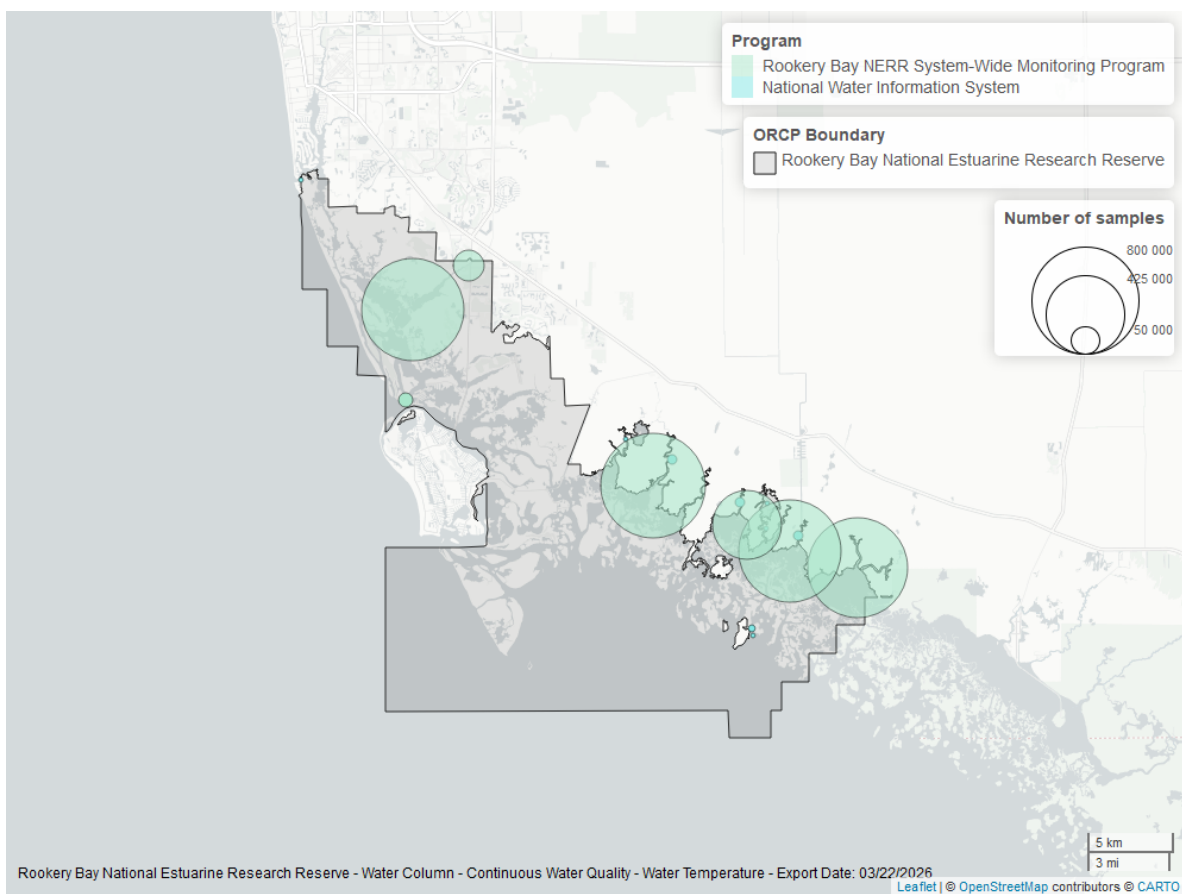


Figure 22: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

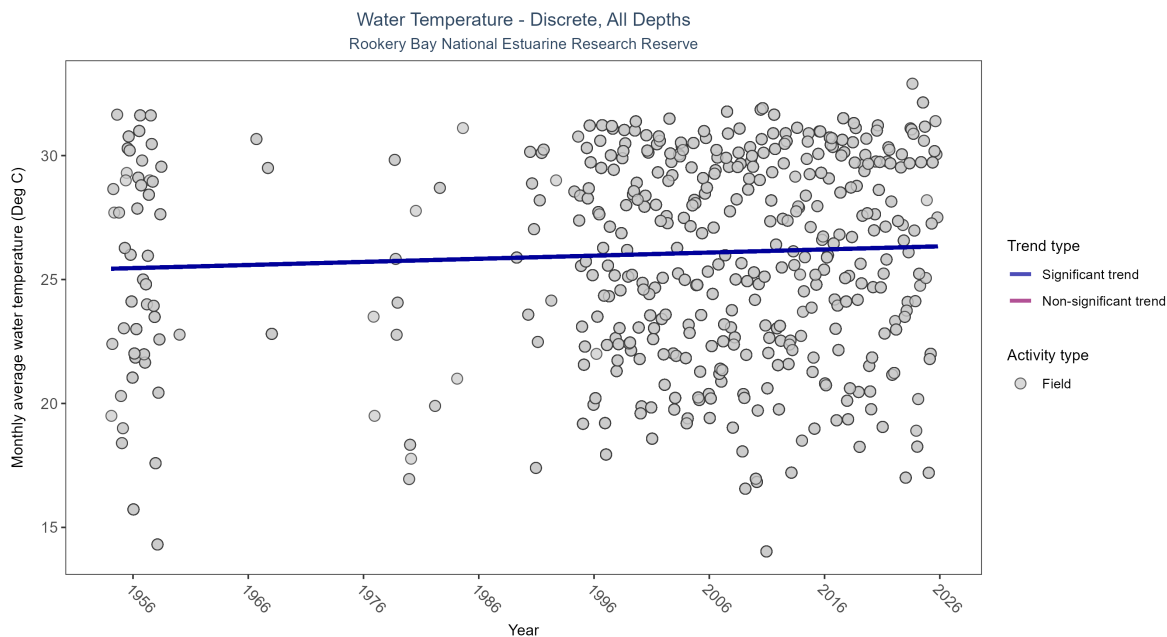


Figure 23: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	15682	51	1954 - 2025	26.63	0.10655	25.43479	0.01256	0.0012

Monthly average water temperature increased by 0.01°C per year.

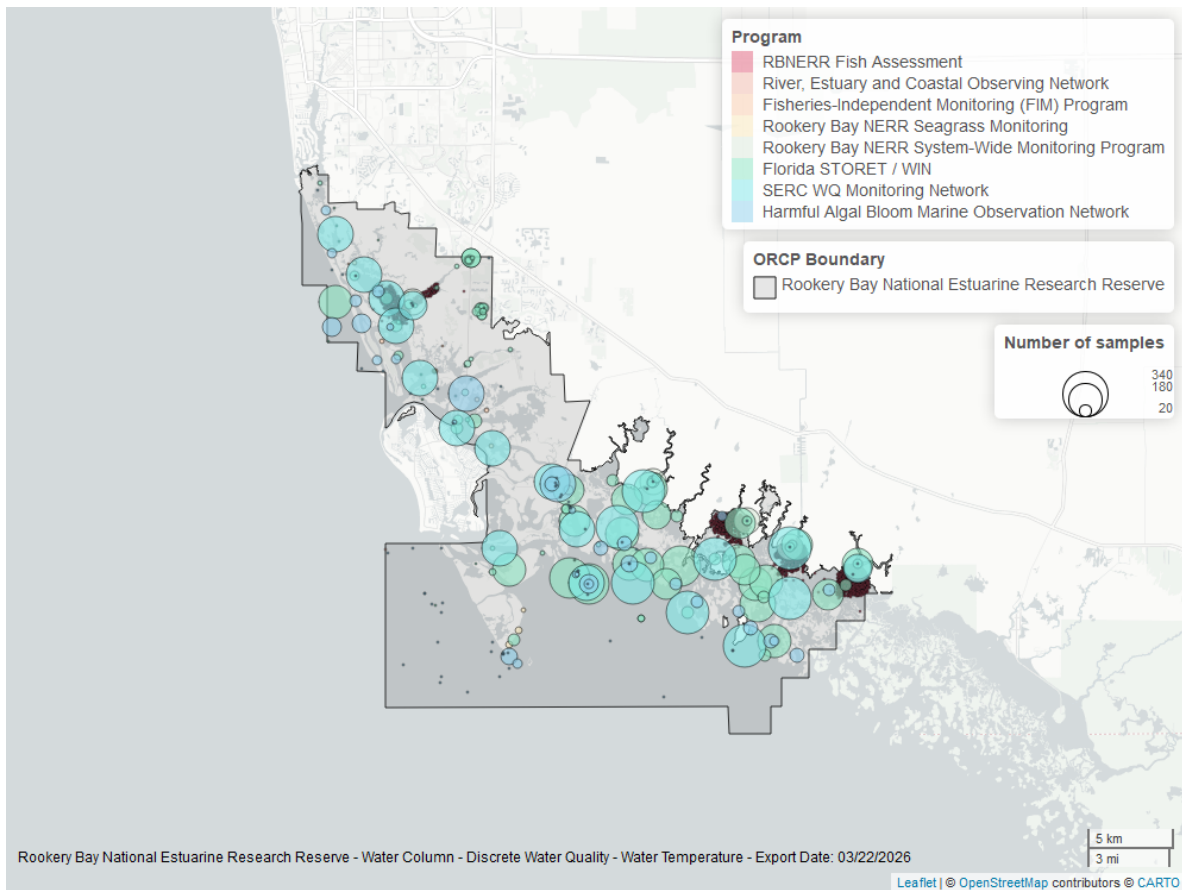


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

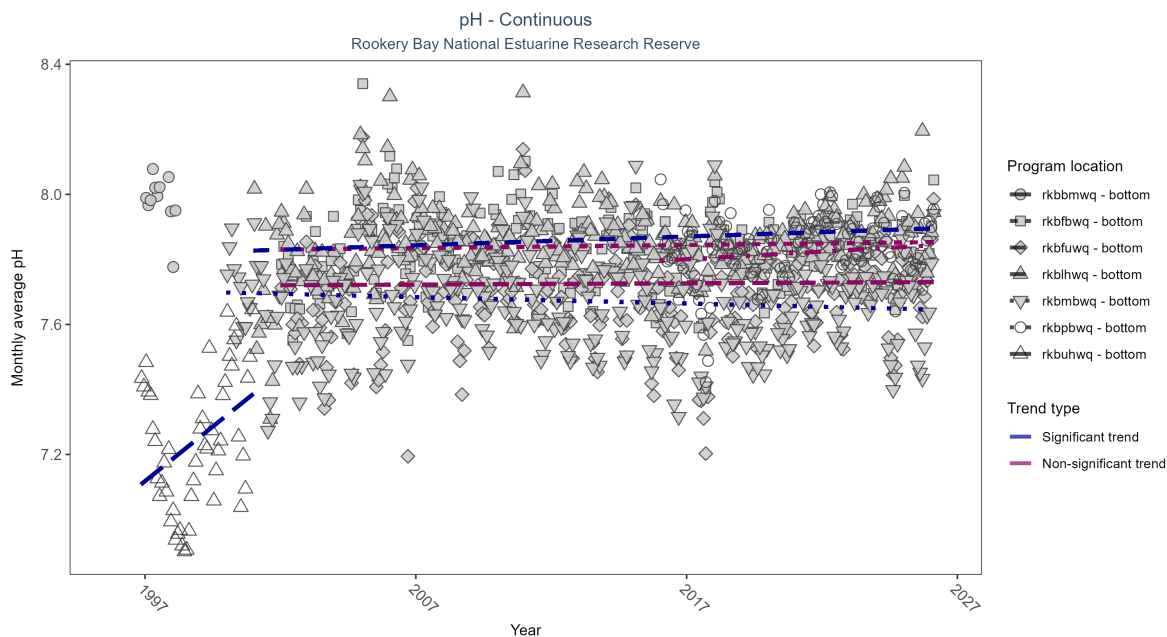


Figure 25: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwq	No significant trend	670684	25	2002 - 2026	7.8	0.05	7.83	0	0.2078
rkbfuwq	No significant trend	689115	25	2002 - 2026	7.7	0.02	7.72	0	0.6096
rkbbmwq	Significantly decreasing trend	727383	27	2000 - 2026	7.7	-0.1	7.7	0	0.0184
rkbpbwq	No significant trend	328113	11	2016 - 2026	7.8	0.13	7.8	0	0.0762
rkbbmwq	Insufficient data to calculate trend	12610	2	1997 - 1998	8.0	-	-	-	-
rkblhwq	Significantly increasing trend	672354	26	2001 - 2026	7.9	0.15	7.83	0	0.0003
rkbuwq	Significantly increasing trend	65814	5	1996 - 2000	7.2	0.37	7.05	0.07	0.0081

At two program locations, monthly average pH increased by less than 0.01 pH units per year at one site and by 0.07 pH units per year at the other. At one program location, monthly average pH decreased by less than 0.01 pH units per year. No detectable change in monthly average pH was observed at three locations. There was insufficient data to fit a model for one location.

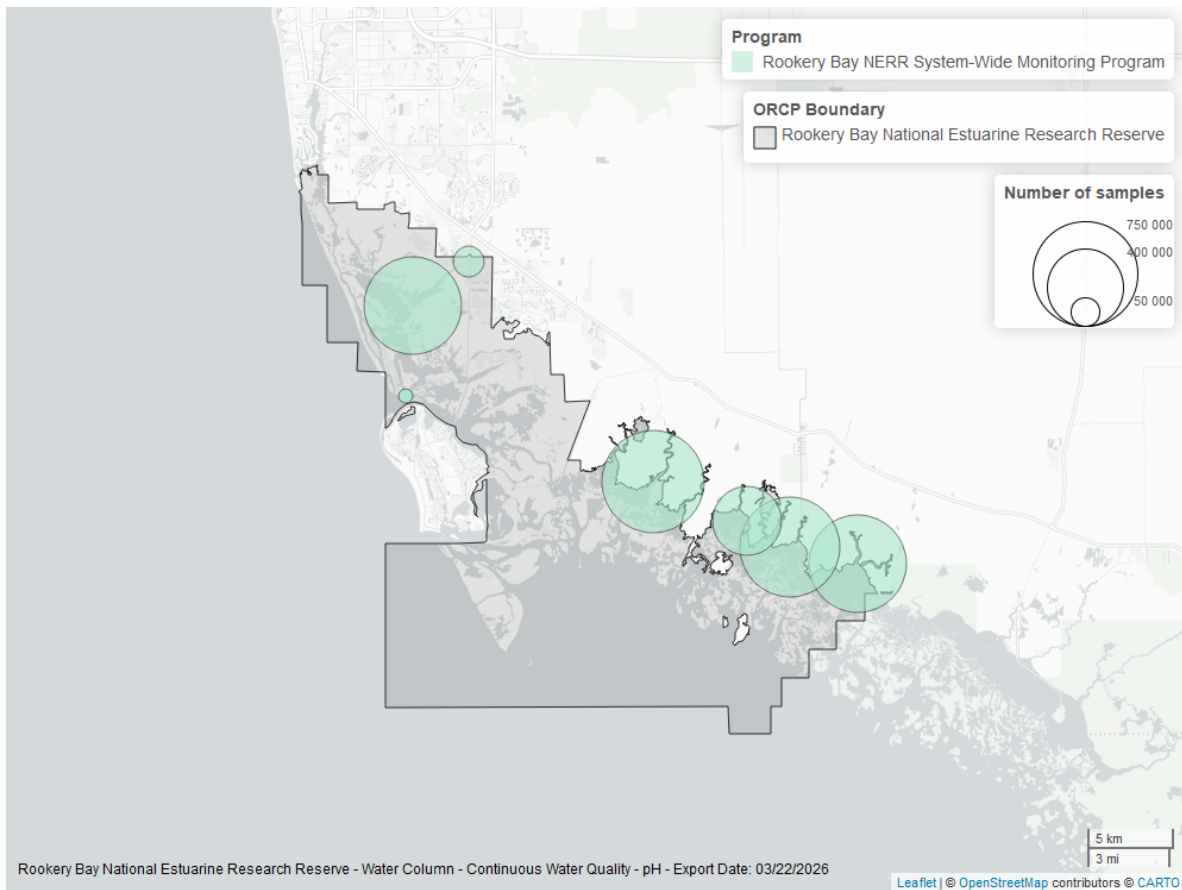


Figure 26: Map showing location of pH continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

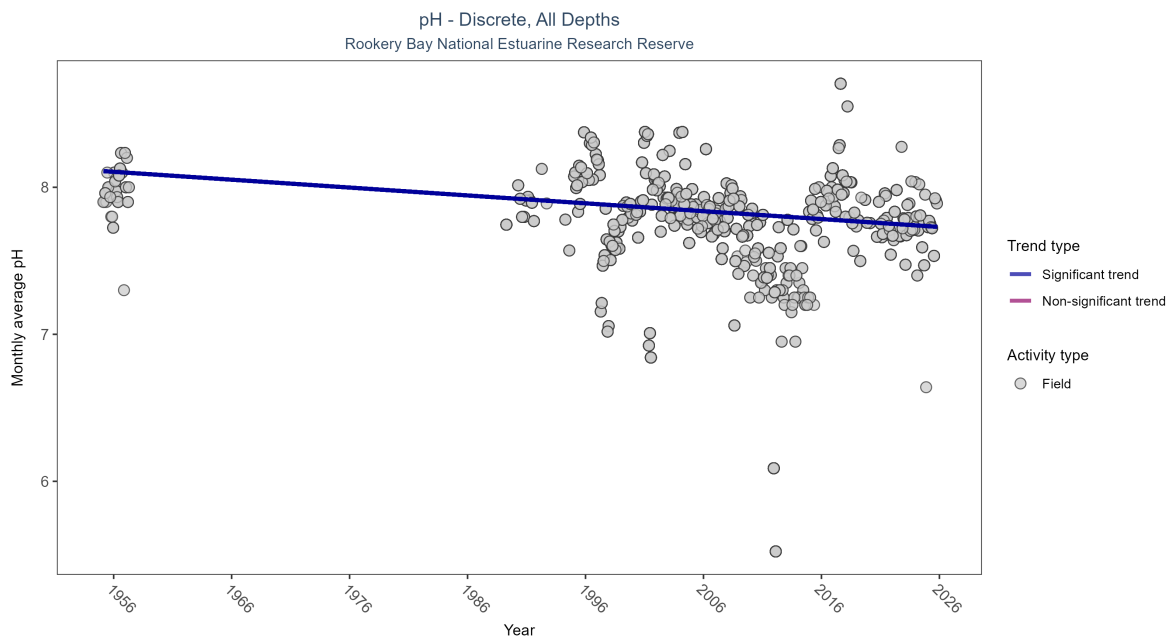


Figure 27: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	6488	39	1955 - 2025	7.88	-0.21625	8.11057	-0.00536	0

Monthly average pH decreased by 0.01 pH units per year.

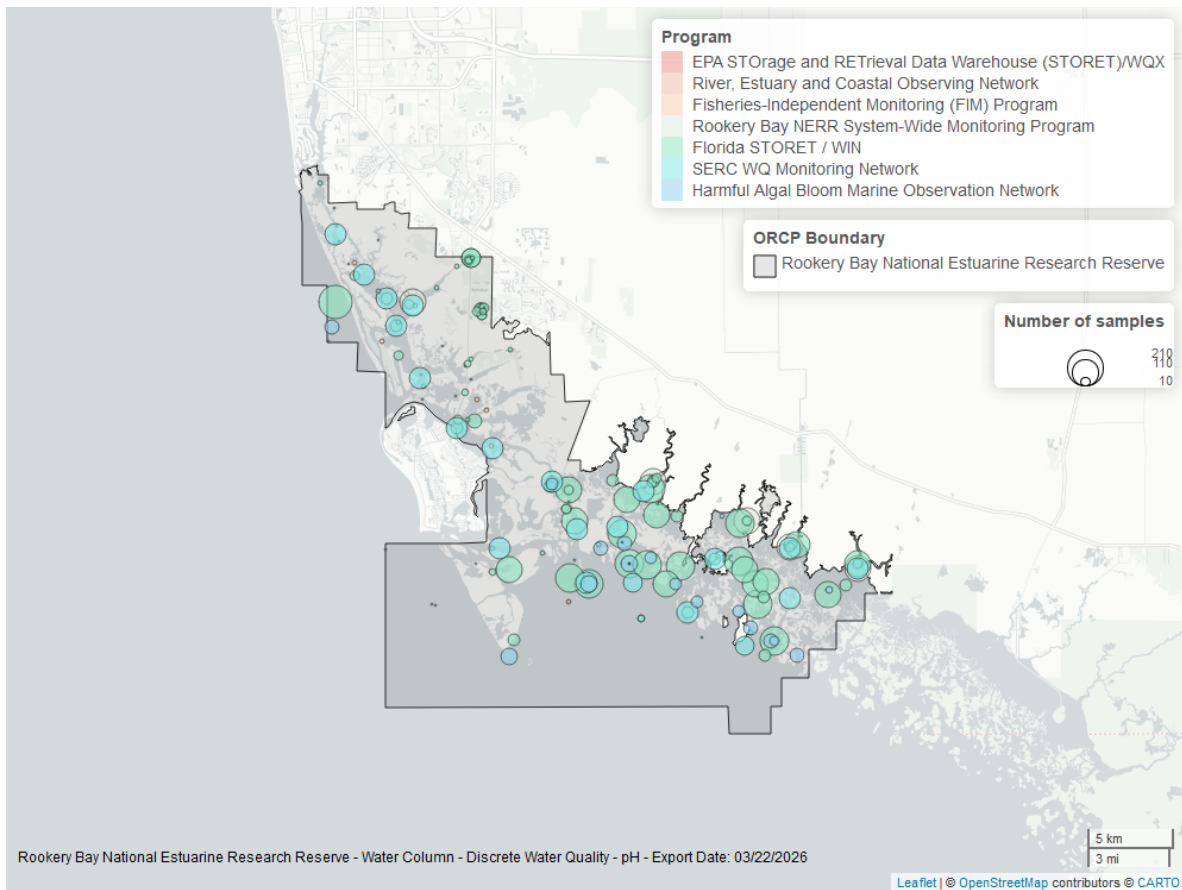


Figure 28: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

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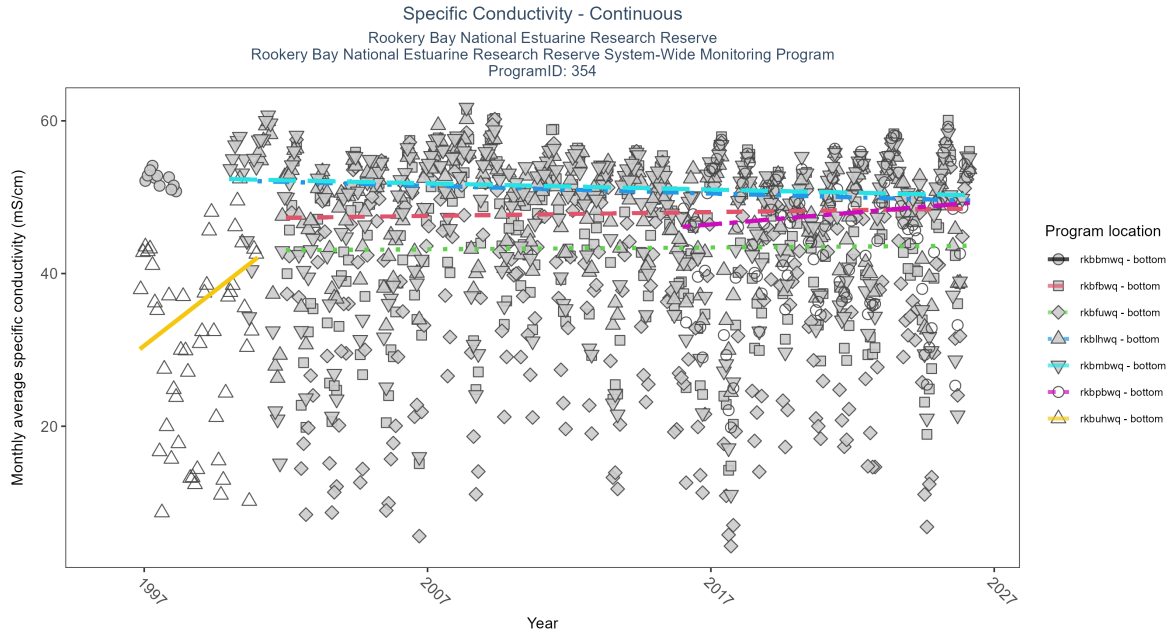


Table 15: Seasonal Kendall-Tau Results for All Stations - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfuwbq	No significant trend	703174	25	2002 - 2026	46.07	0.04	47.3	0.05	0.3304
rkbfuwbq	No significant trend	717061	25	2002 - 2026	41.17	0.01	43.06	0.02	0.7747
rkbbmwq	Significantly decreasing trend	736129	27	2000 - 2026	50.83	-0.09	52.37	-0.08	0.0197
rkbpbwq	No significant trend	333114	11	2016 - 2026	47.03	0.13	46.12	0.31	0.068
rkbbmwq	Insufficient data to calculate trend	12256	2	1997 - 1998	52.41	-	-	-	-
rkblhwq	Significantly decreasing trend	700390	26	2001 - 2026	50.59	-0.12	52.11	-0.1	0.003
rkbuwbq	No significant trend	68971	5	1996 - 2000	33.92	0.29	27.67	2.88	0.0552

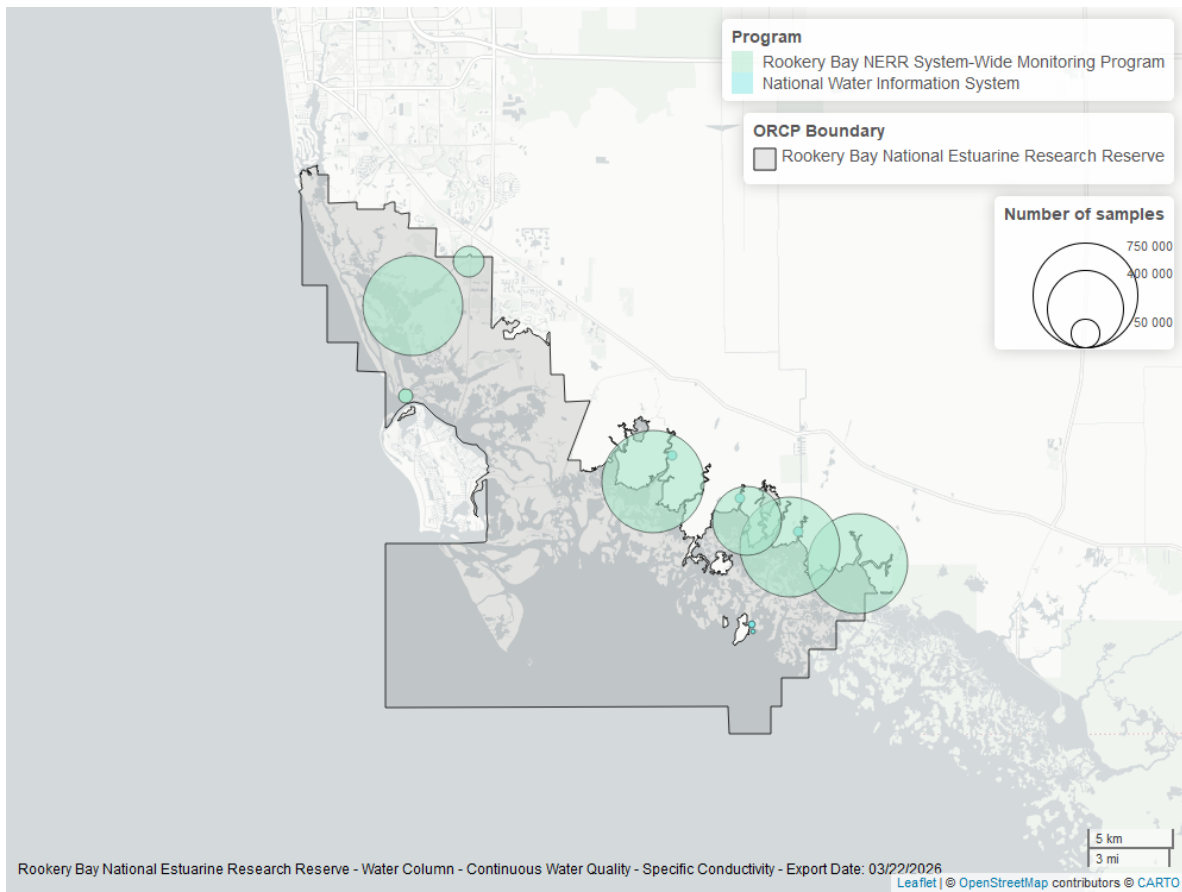


Figure 29: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

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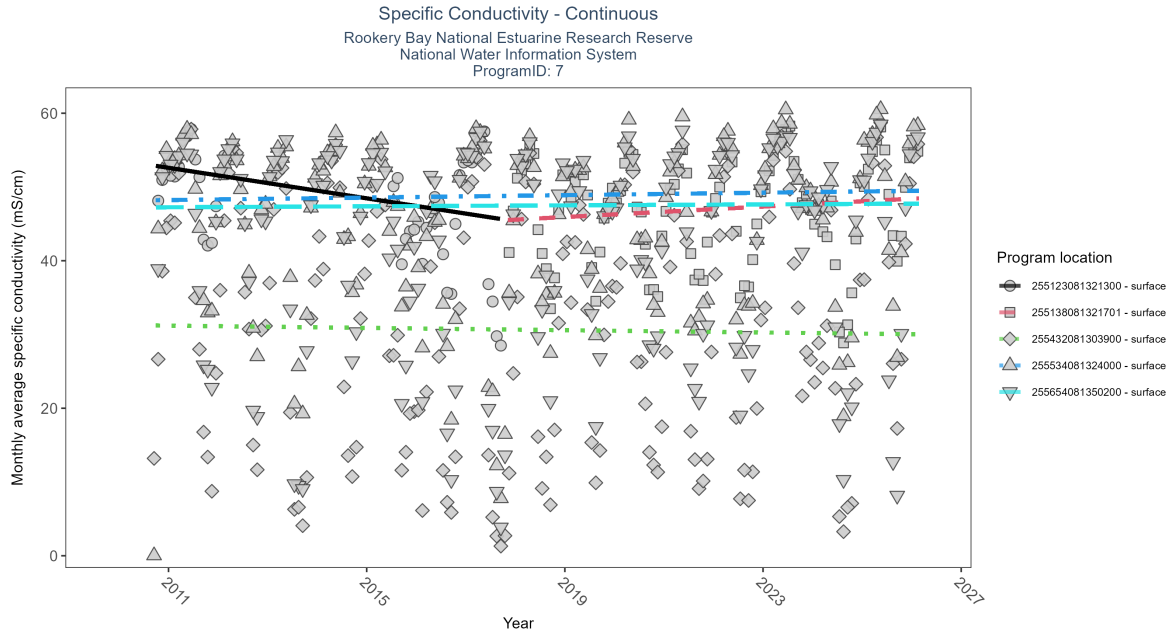


Table 16: Seasonal Kendall-Tau Results for All Stations - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	No significant trend	1295	6	2010 - 2017	49.1	-0.20	53.63	-1.03	0.2034
255138081321701	Significantly increasing trend	2986	10	2017 - 2026	48.3	0.18	45.21	0.36	0.0248
255432081303900	No significant trend	5512	17	2010 - 2026	34.4	-0.03	31.28	-0.08	0.6149
255534081324000	No significant trend	5509	17	2010 - 2026	48.6	0.04	48.14	0.08	0.4303
255654081350200	No significant trend	5547	17	2010 - 2026	48.6	0.03	47.21	0.03	0.5649

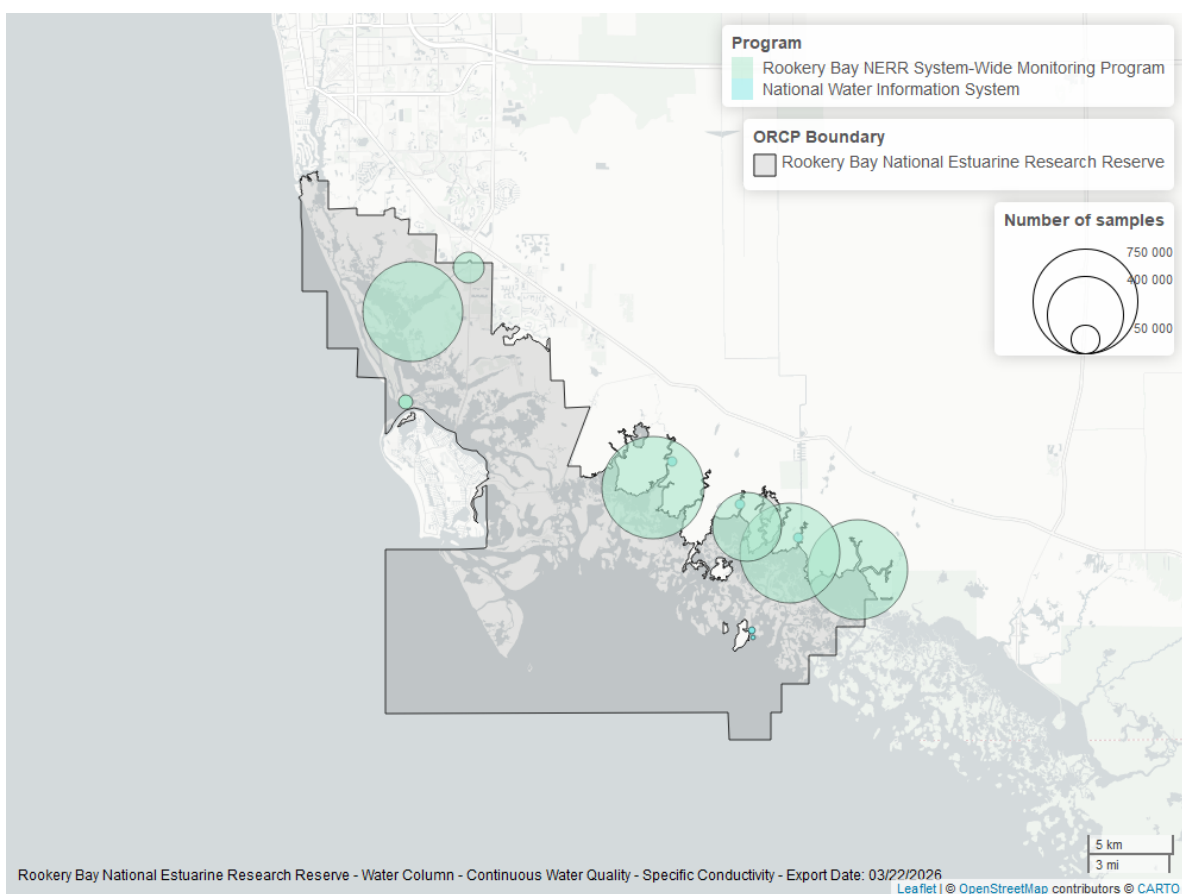


Figure 30: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

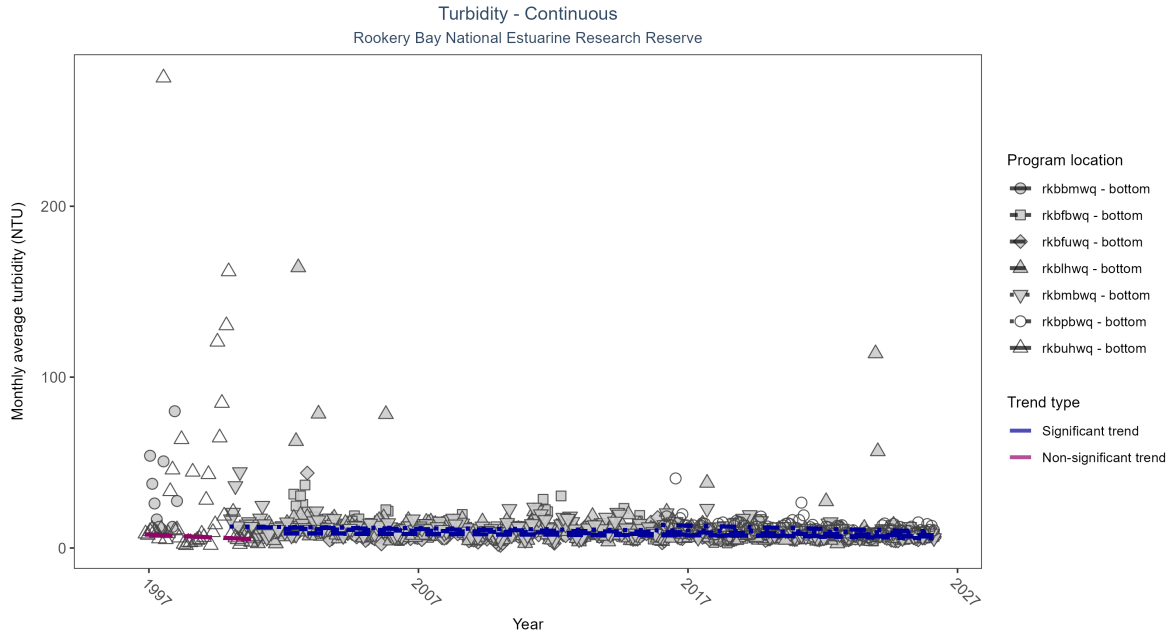


Figure 31: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 17: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwq	Significantly decreasing trend	682915	25	2002 - 2026	7	-0.4	11.3	-0.23	0
rkbfuwq	Significantly decreasing trend	657778	25	2002 - 2026	6	-0.19	8.57	-0.08	0
rkmbmwq	Significantly decreasing trend	714237	27	2000 - 2026	9	-0.24	12.56	-0.14	0
rkbpbwq	Significantly decreasing trend	331299	11	2016 - 2026	9	-0.31	13.47	-0.37	0
rkbbmwq	Insufficient data to calculate trend	10654	2	1997 - 1998	11	-	-	-	-
rkblhwq	Significantly decreasing trend	647527	26	2001 - 2026	8	-0.28	12.22	-0.19	0
rkbuwq	No significant trend	61608	5	1996 - 2000	5	-0.11	8.57	-0.7	0.5228

At five program locations, monthly average turbidity decreased between 0.08 and 0.37 NTU per year. No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for one location.

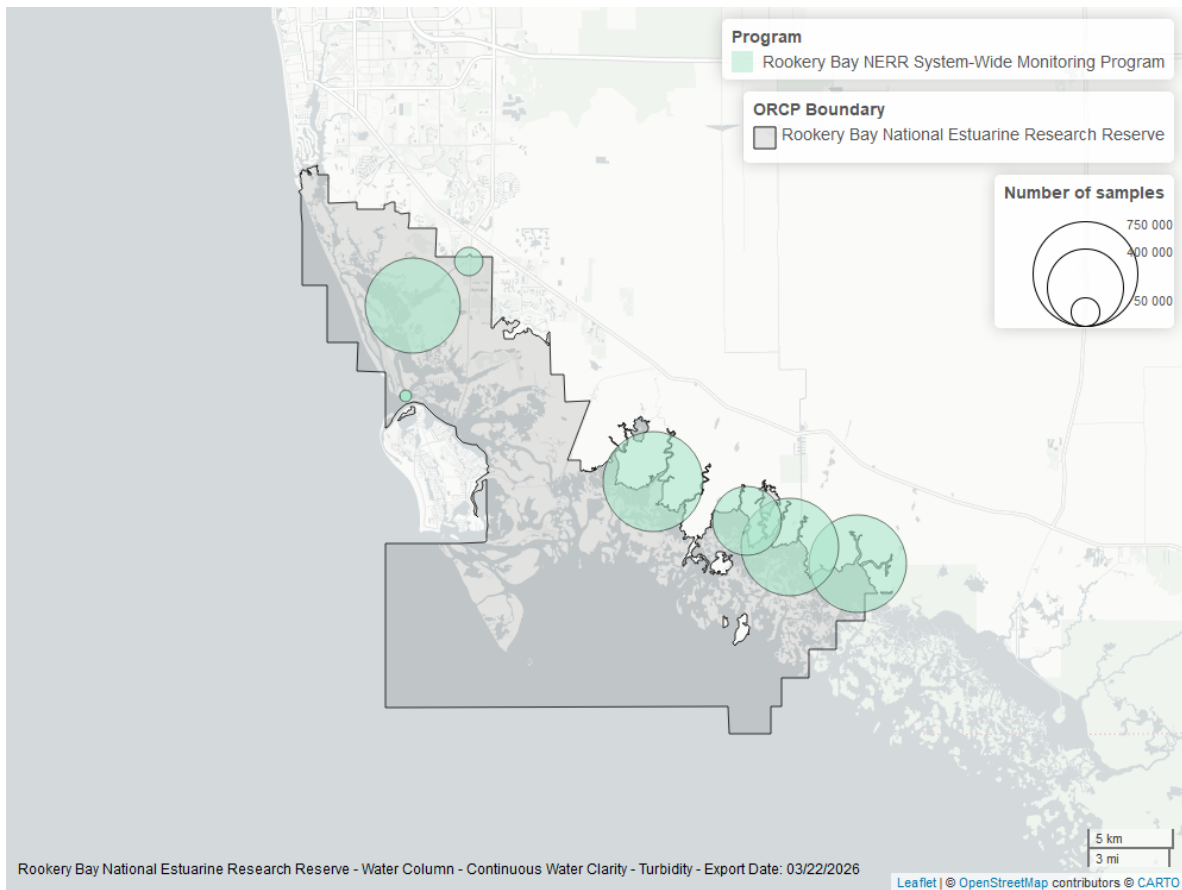


Figure 32: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

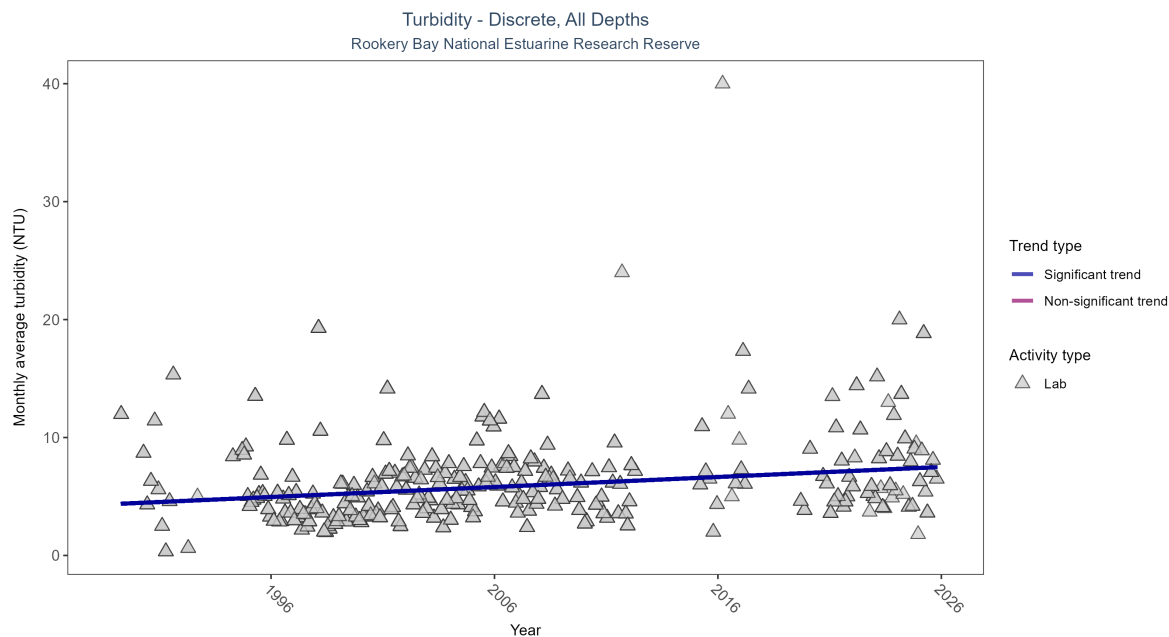


Figure 33: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	6438	33	1989 - 2025	4.3975	0.20684	4.35994	0.0849	0

Monthly average turbidity increased by 0.08 NTU per year, indicating a decrease in water clarity.

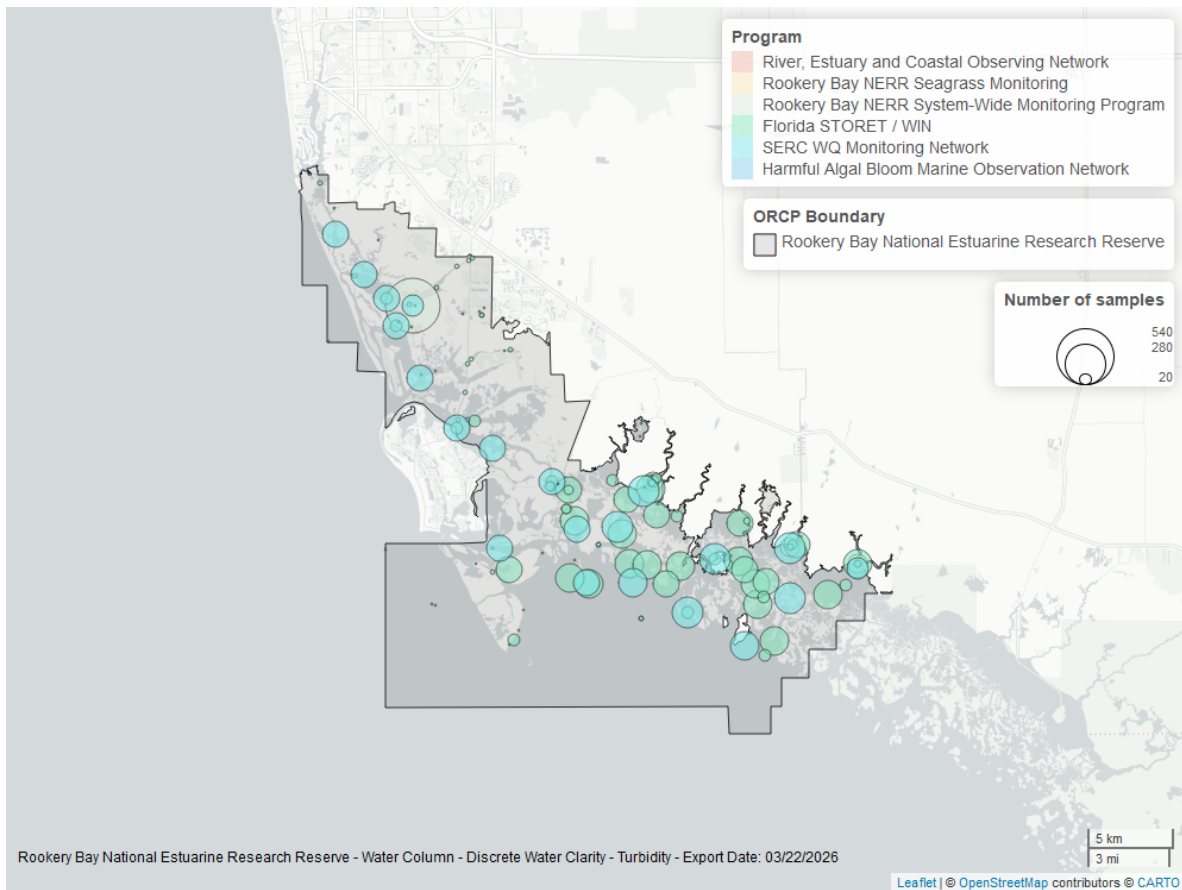


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

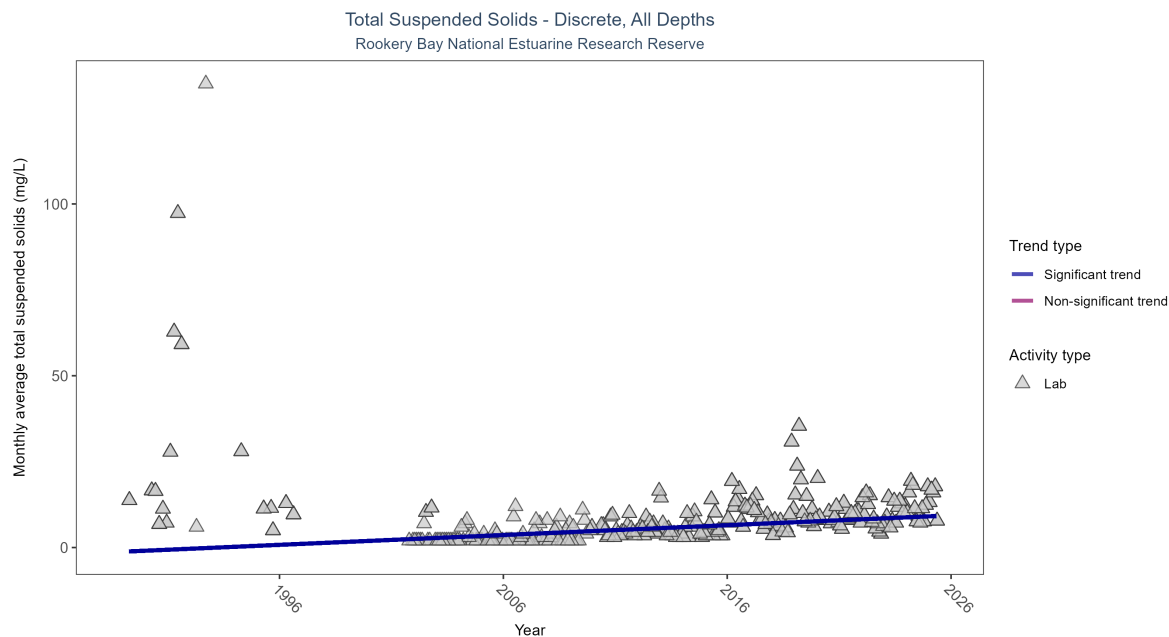


Figure 35: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 19: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	906	32	1989 - 2025	8	0.38553	-1.21736	0.28571	0

Monthly average total suspended solids increased by 0.29 mg/L per year, indicating a decrease in water clarity.

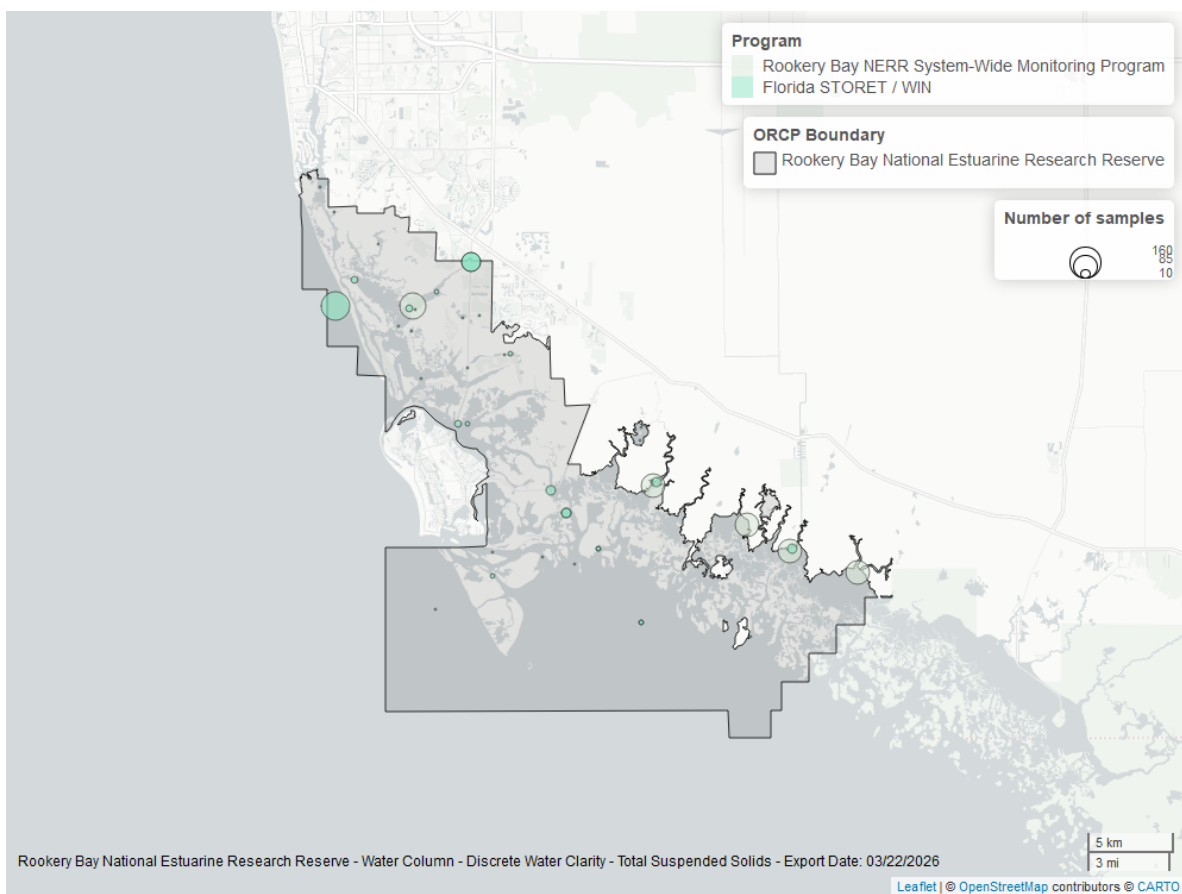


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

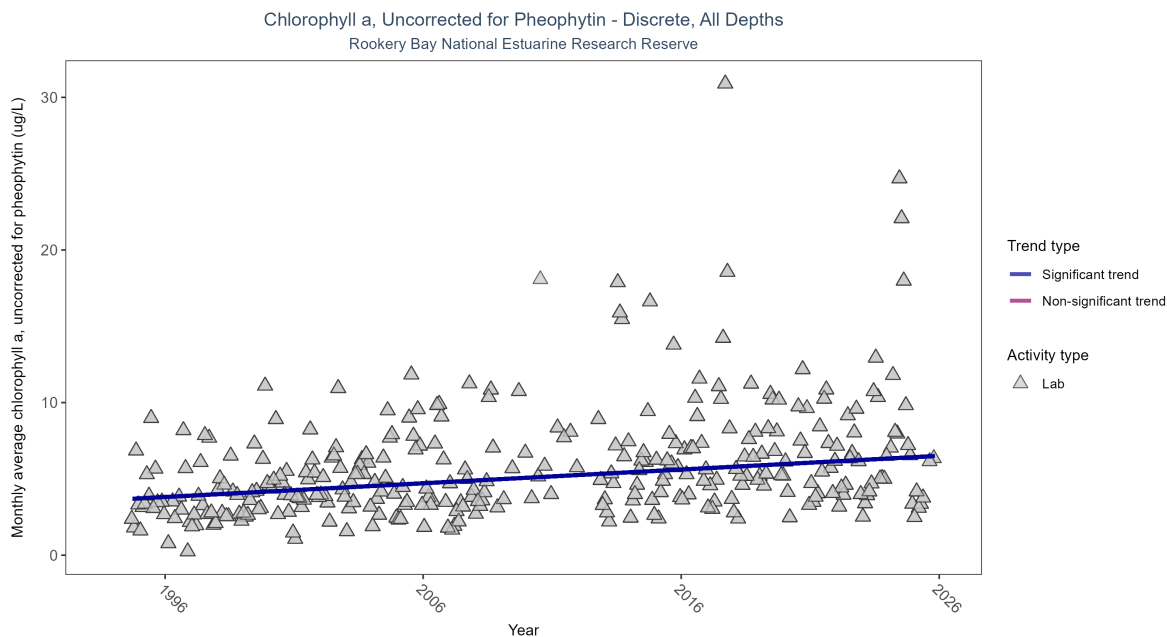


Figure 37: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	5487	32	1994 - 2025	4.4	0.31615	3.63368	0.08998	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.09 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

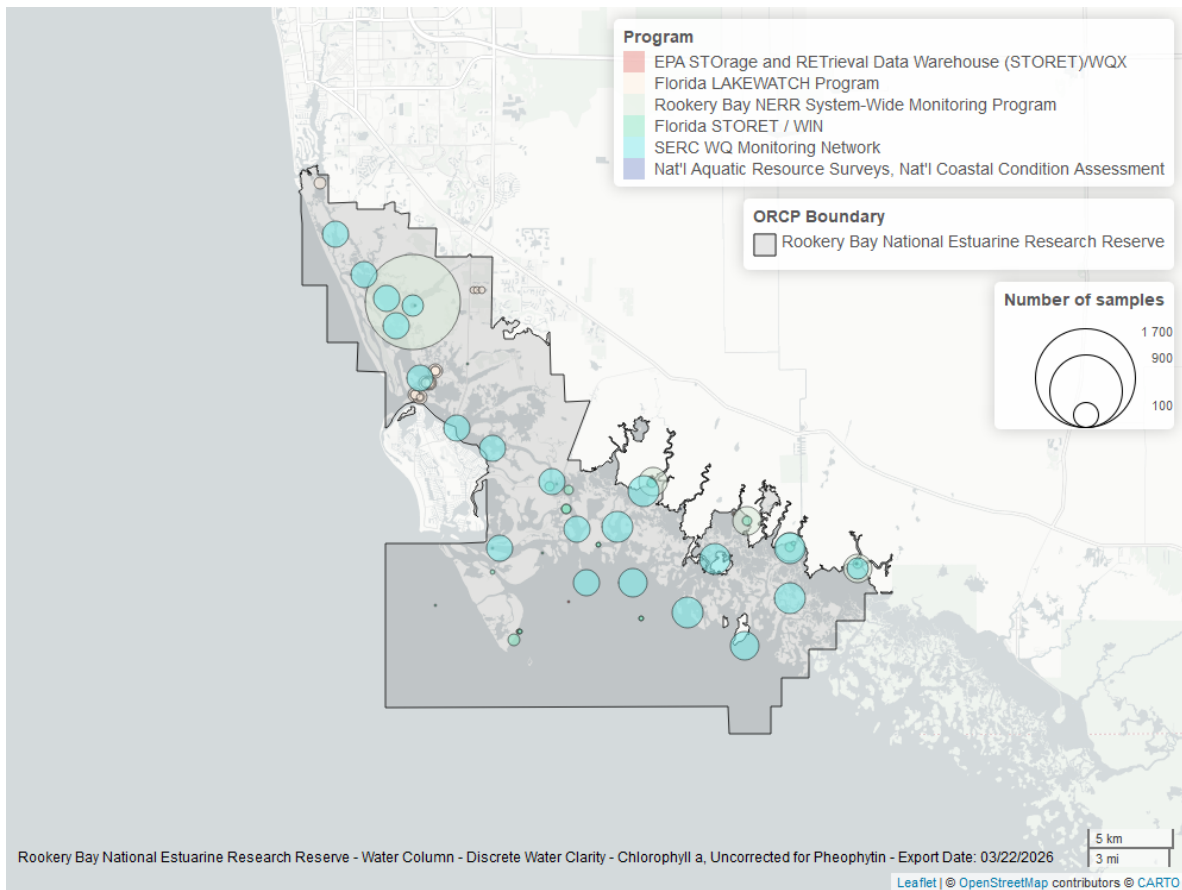


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

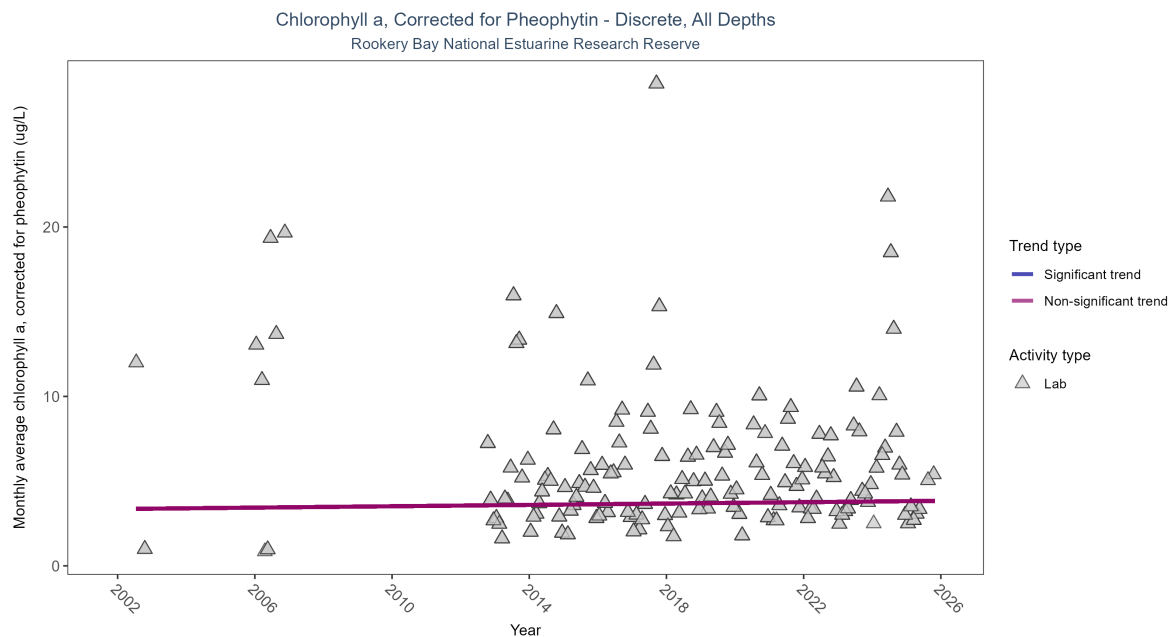


Figure 39: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 21: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	2391	16	2002 - 2025	4.1	0.04598	3.35585	0.02004	0.4535

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2025.

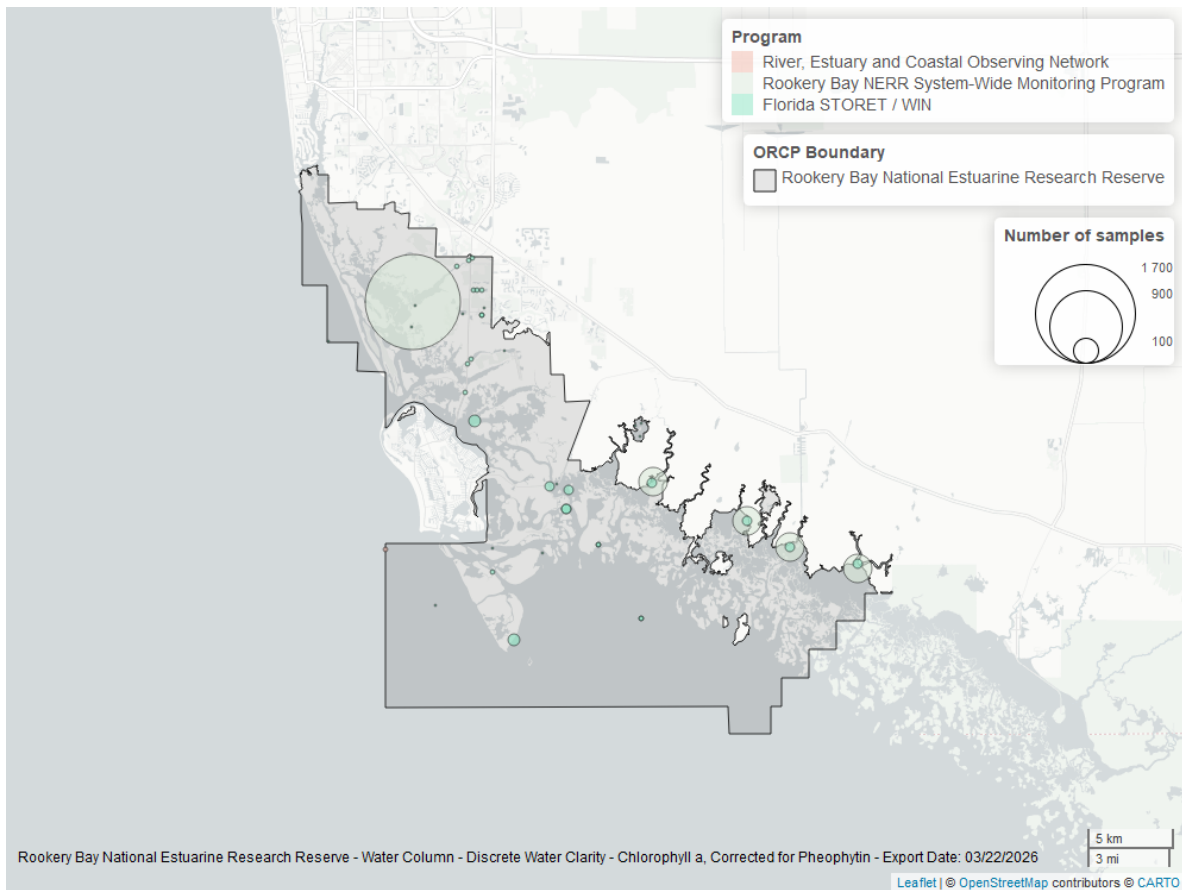


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

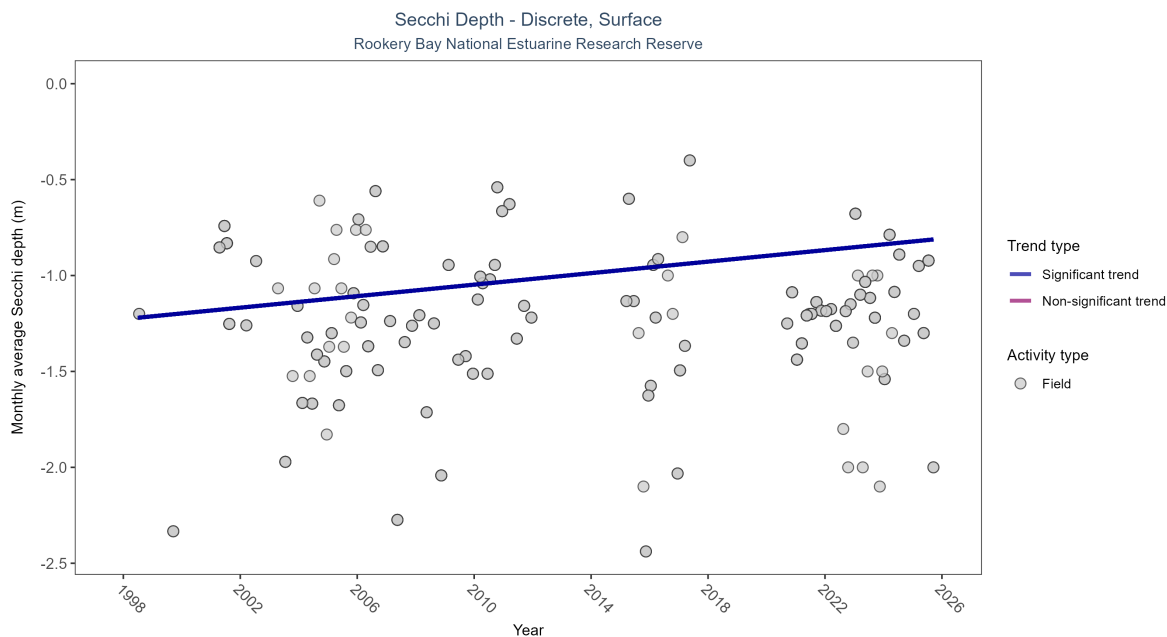


Figure 41: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 22: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	740	22	1998 - 2025	-1.06681	0.24016	-1.22762	0.015	0.0003

Monthly average Secchi depth became shallower by 0.01 m per year, indicating a decrease in water clarity.

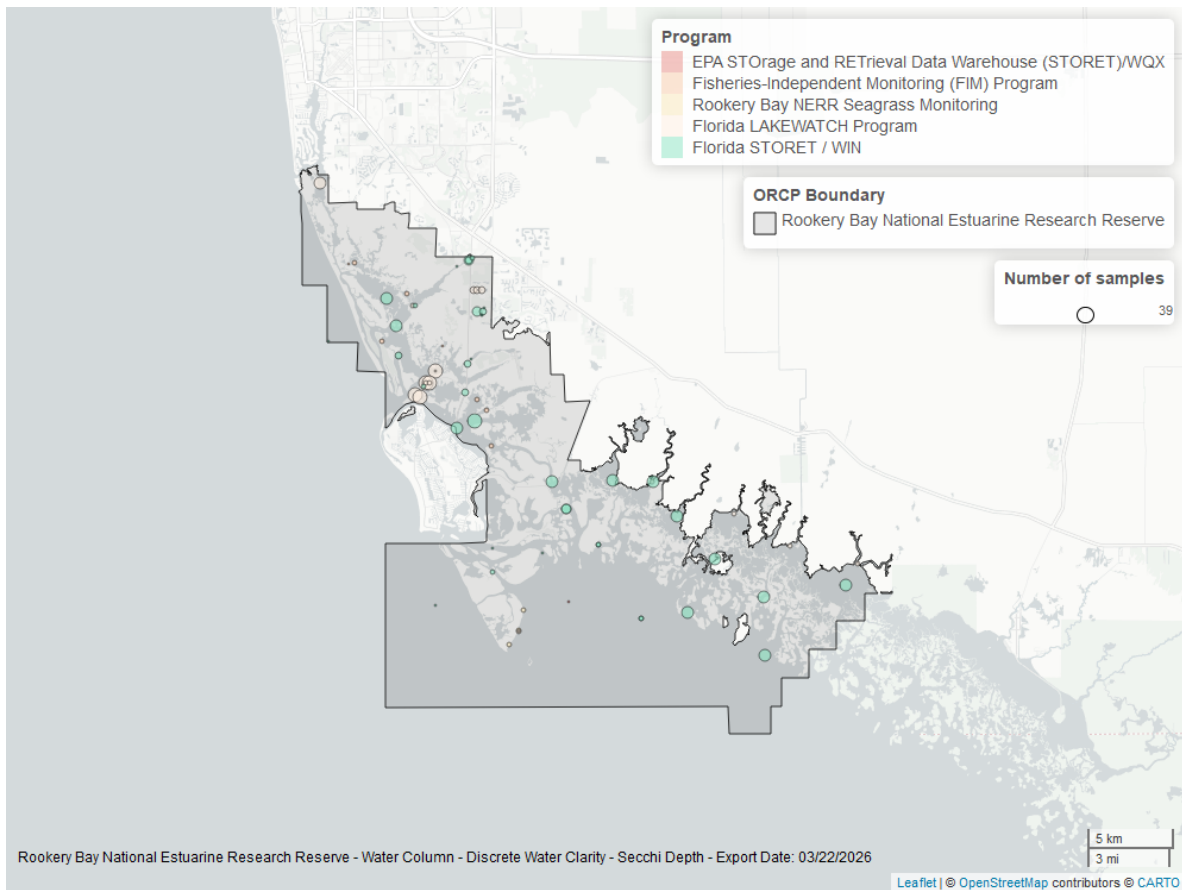


Figure 42: Map showing location of discrete water quality sampling locations within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.